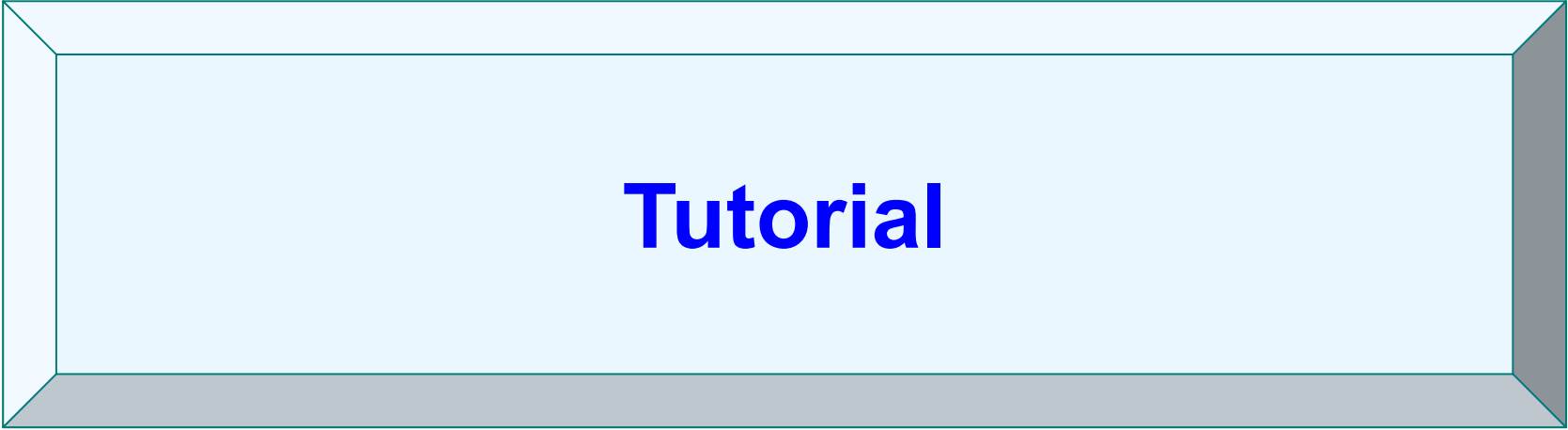
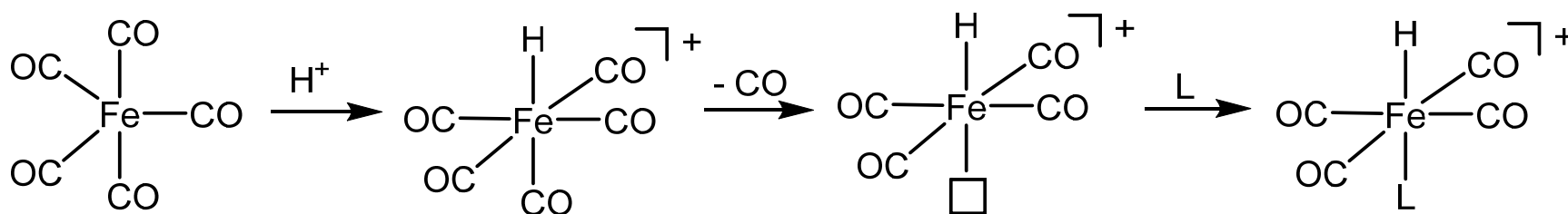




Tutorial

1. Shot Questions

$\text{Fe}(\text{CO})_5$ loses CO very slowly, but in the presence of an acid, substitution is greatly accelerated. Suggest possible explanations.

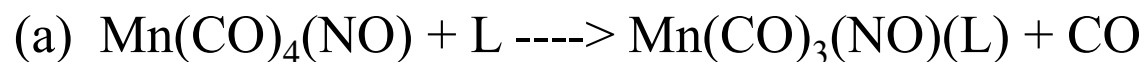


Answer: Protonation at the metal (always allowed even for 18e complexes) should introduce a cationic charge, that should not only weaken M–CO bonding (decrease the back donation from M to CO) but also put a high-trans-effect H ligand on the metal.

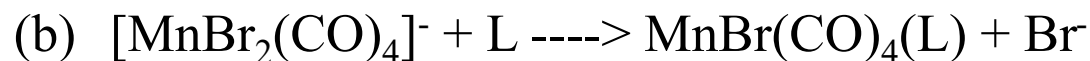
Multiple Choices

For each question, select the most appropriate answer.

Q1. Which of the following substitution reactions may undergo through a associative mechanism?



18e for linear NO; 16e for bent NO



18e, dissociation of Br⁻ first

(c) Both of them

(d) none of them

a)

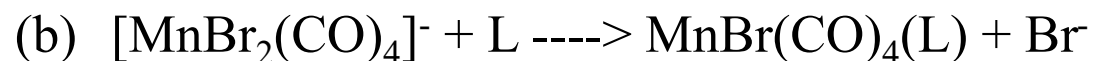
2. Multiple Choices

For each question, select the most appropriate answer.

Q2. Which of the following substitution reactions may undergo through a associative mechanism?



$\eta^5\text{-Cp}$ may rearrange to $\eta^3\text{-Cp}$



18e, dissociation of Br first

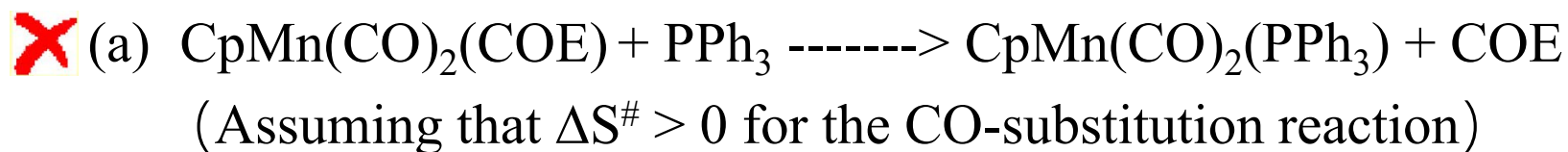
(c) both of them

(d) none of them

Multiple Choices

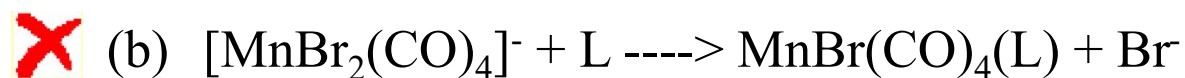
For each question, select the most appropriate answer.

Q3. Which of the following substitution reactions may undergo through a associative mechanism?



Since $\Delta S^\ddagger$ is positive, it is a dissociative substitution.

(Although it is possible that $\eta^5\text{-Cp}$ may rearrange to $\eta^3\text{-Cp}$);



18e, dissociation of Br⁻ first

(c) both of them

(d) none of them 

Select the most appropriate answer.

Q4. Which of the following **substitution reactions** would you expect to have the **highest reaction rate**?



Answer: (b)

CFSE consideration (晶体场稳定化能)

Ni(0), d^{10} , Th, no CFSE. Ni-L bond is weaker;

**Ni-NCCH₃ is
weaker than Ni-CO**

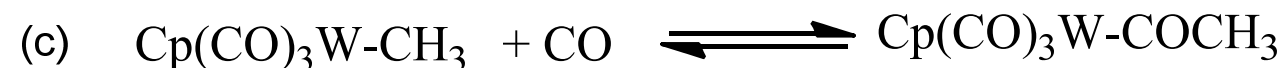
Mo(0), Cr(0), d^6 , Oh, big CFSE. M-L bond is strong

**Mo-CO is stronger
than Cr-CO**

**** The substitution reactions of d^3 , d^6 (low spin) octahedral complexes are usually very slow.***

Select the most appropriate answer.

Q4. Which of the following reactions would you expect to have the largest equilibrium constant K_{eq} ?

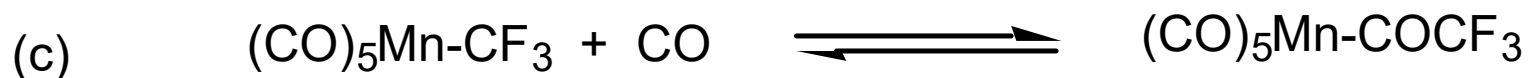
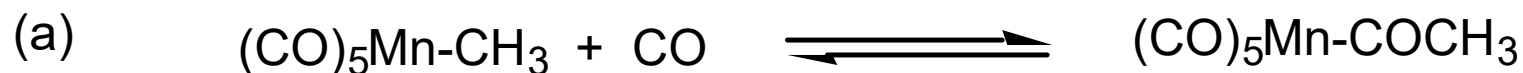


Bond strength of M-R

a)

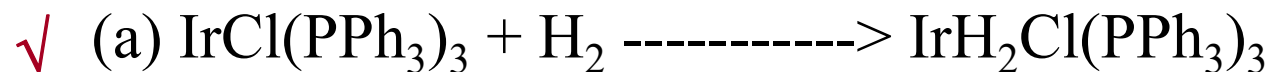
Select the most appropriate answer.

Q4. Which of the following reactions would you expect to have the **smallest equilibrium constant K_{eq}** ?



d)

Q5. How many of the following oxidative addition reactions is/are likely to proceed through a concerted mechanism?



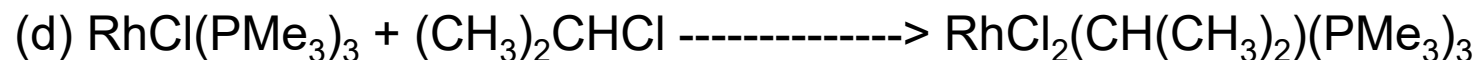
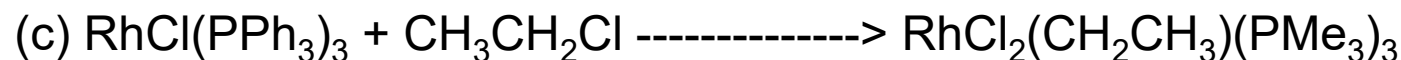
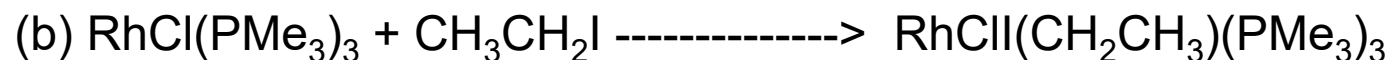
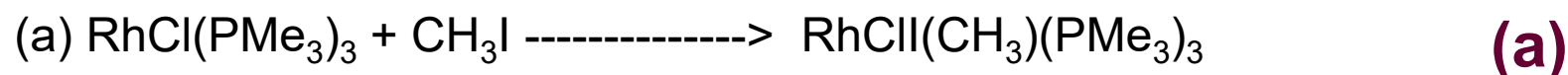
✓ (e) two of them (f) three of them (g) all of them

(h) none of them

e)

Select the most appropriate answer.

Q6. Which of the following oxidative addition reactions would you expect to occur **most readily**? Assuming a non-concerted, $\text{S}_{\text{N}}2$ pathway is involved in the reactions.

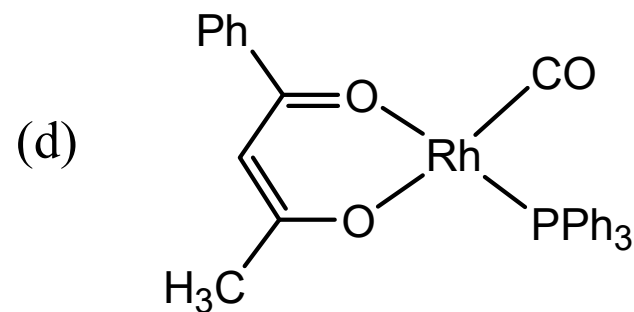
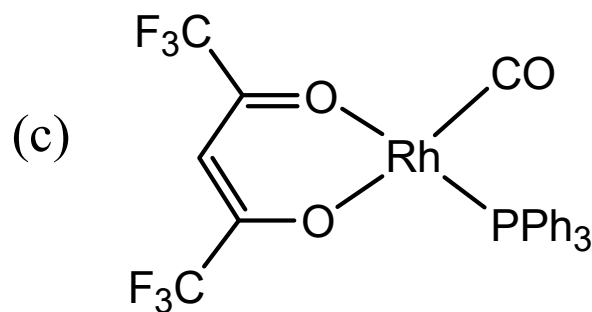
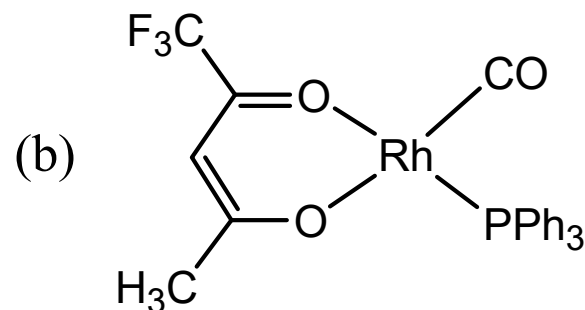
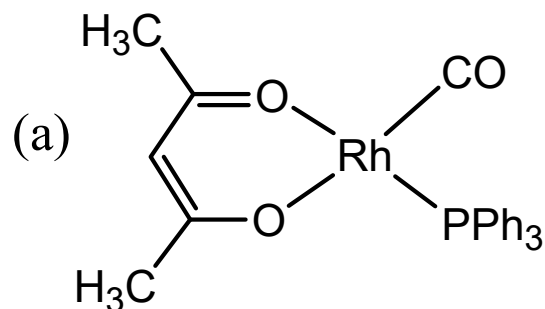


$\text{S}_{\text{N}}2$ reactions:

e- richness of metal complex
Steric effect;
Leaving ability.

Review Question: Multiple Choices

Q7. Which of the following complexes would you expect to undergo an oxidative addition reaction with CH_3I at the **lowest reaction rate**?



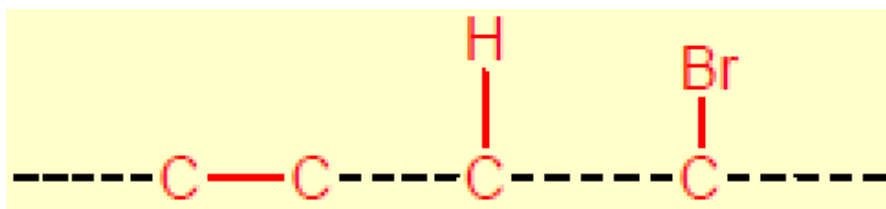
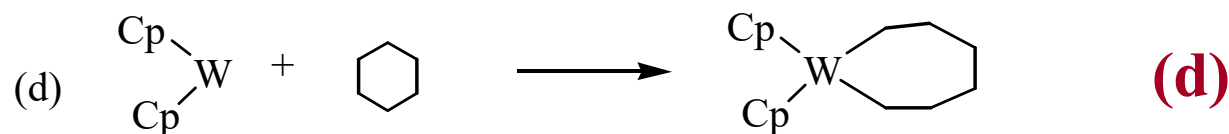
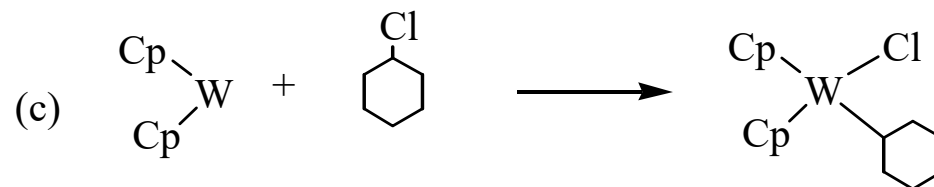
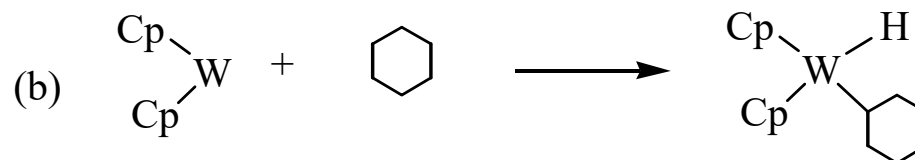
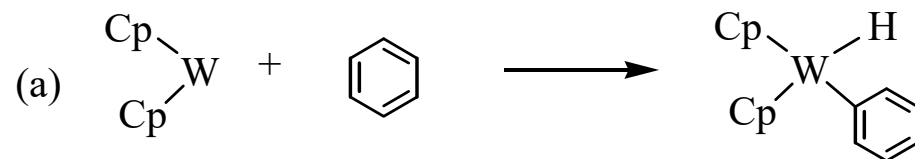
C)

e- richness of metal complex;

The acac ligand in c is less e-rich than others;

Select the most appropriate answer.

Q8. Which of the following reactions is **least likely** to occur?



Select the most appropriate answer.

Q9. Reaction of $\text{Cp}_2\text{ZrCl}(\text{Me})$ with PhSiH_3 produces $\text{Cp}_2\text{ZrCl}(\text{SiH}_2\text{Ph})$ and CH_4 . Which of the following species is the likely intermediate?

✓ (a) $\text{Cp}_2\text{ZrCl}(\text{Me})(\eta^2\text{-SiH}_3\text{Ph})$
18e, Zr(IV)

✗ (b) $\text{Cp}_2\text{ZrClH}(\text{Me})(\text{SiH}_2\text{Ph})$
Zr(VI)

✗ (c) $\text{Cp}_2\text{ZrCl}(\text{Me})(\text{Ph})(\text{SiH}_3)$
Zr(VI) OA of Si-Ph

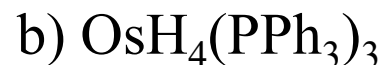
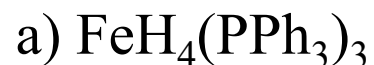
✗ (d) $\text{Cp}_2\text{ZrCl}(\text{Me})(\eta^2\text{-SiH}_3\text{Ph})_2$
20e, Zr(IV)

$\text{Cp}_2\text{ZrCl}(\text{Me})$, 16e, Zr(IV) $\rightleftharpoons$ σ -bond metathesis

Answer: (a)

Select the most appropriate answer.

Q10. $\text{RuH}_4(\text{PPh}_3)_3$ is known to contain a dihydrogen ligand [therefore it should be formulated as $\text{RuH}_2(\text{H}_2)(\text{PPh}_3)_3$]. Which of the following compounds is **least likely to contain a dihydrogen** ligand?

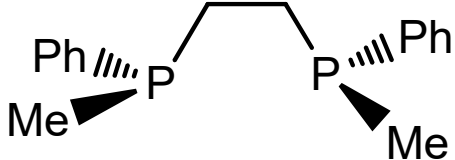
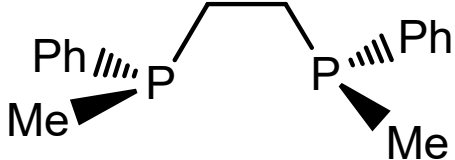
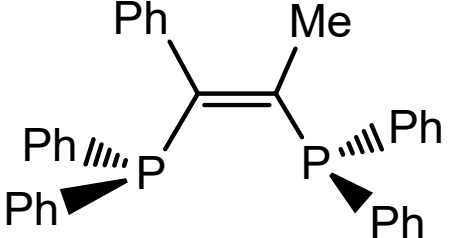
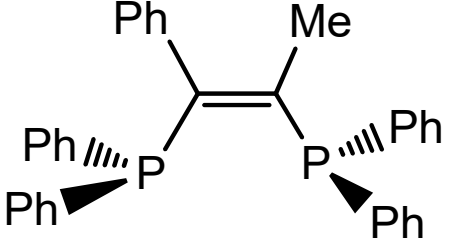
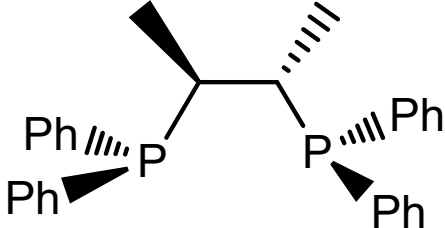
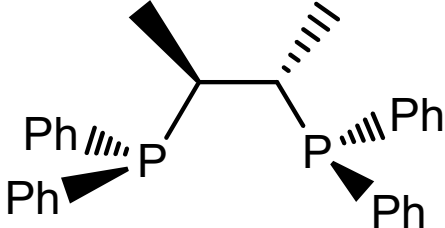
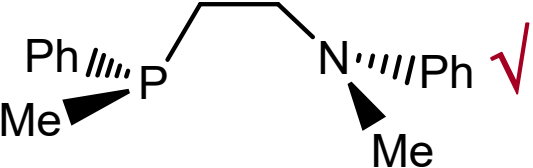
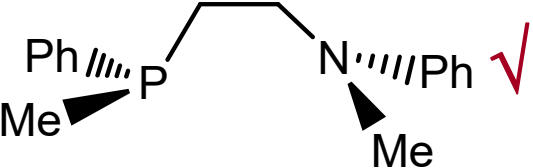


(b)

Multiple Choices

For each question, select the most appropriate answer.

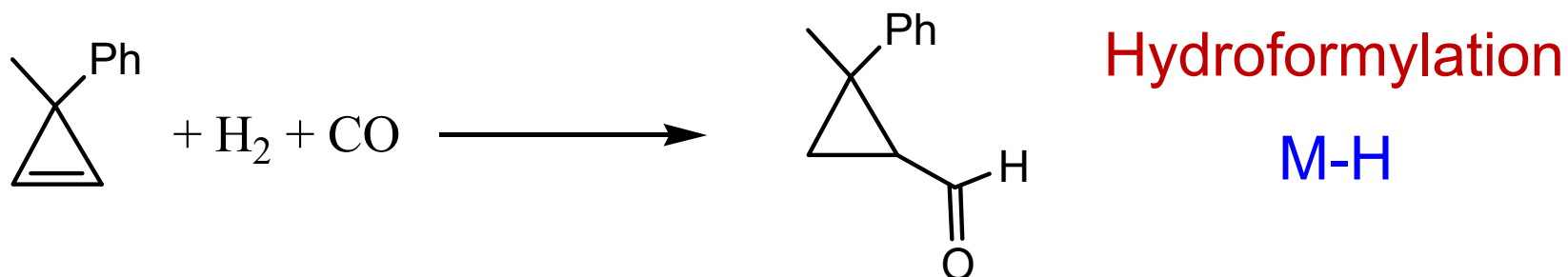
Q11. How many of the following ligands could be used for asymmetric catalytic hydrogenation of $\text{C}_6\text{H}_5\text{C}(\text{Me})=\text{CHCONMe}_2$?

- (a)  (a) 
- (b)  (b) 
- (c)  (c) 
- (d)  (d) 
- (e) two of them
- (f) three of them

Answer: (e)

Select the most appropriate answer.

Q12. How many of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



✓ (a) $\text{HRh(PMe}_3)_4$

✓ (c) $\text{Co}_2(\text{CO})_8$

(e) two of them

(g) all of them

✓ (b) RhH(CO)_4

✗ (d) $\text{PdCl}_2/\text{CuCl}_2$

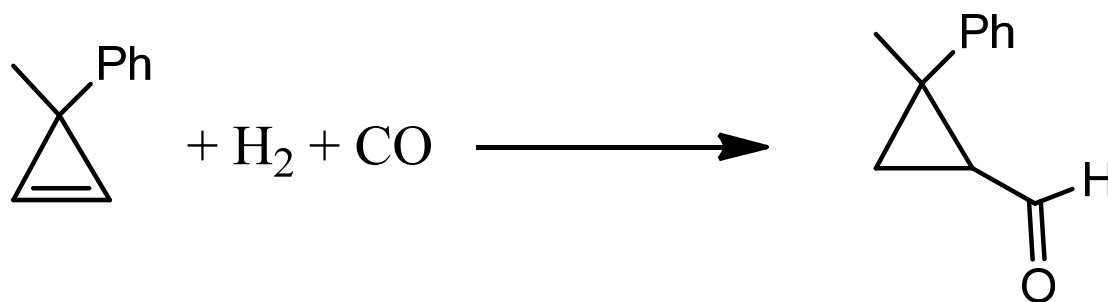
(f) three of them

(h) none of them



Answer: (f)

Q13. How many of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



✓ (a) $\text{HRh(PMe}_3)_4$

(b) RhH(CO)_4 ✓

✓ (c) $\text{Co}_2(\text{CO})_8$

(d) $\text{PtHCl(PPh}_3)_2$ ✓

(e) two of them

(f) three of them

(g) all of them

(h) none of them


Answer: (g)

Select the most appropriate answer.

Q14. How many of the following complexes could be used as the catalyst or catalytic precursor for polymerization of $\text{PhC}\equiv\text{CH}$?

✓ (a) $\text{RhH}(\text{PCy}_3)_3$

✓ (b) $(\text{RO})_3\text{Mo}\equiv\text{CCMe}_3$

✓ (c) $\text{Br}_2(\text{RO})_2\text{Mo}=\text{CHPh}$

(d) Cp_2Fe ✗

(e) two of them

(f) three of them



(g) all of them

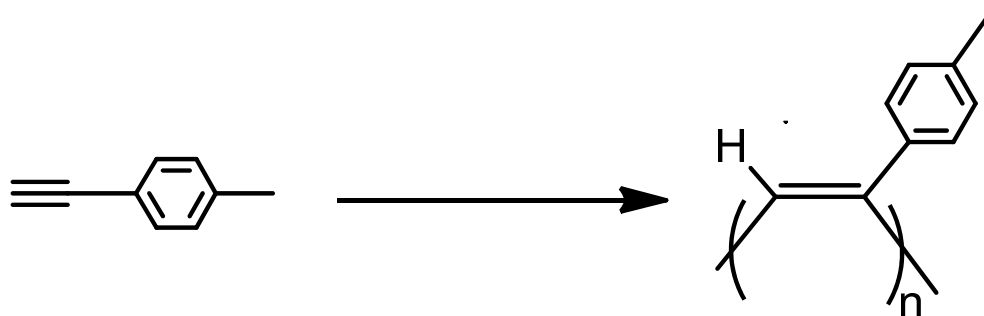
(h) none of them

M-R, M=C, M $\equiv$ C

Answer: (f)

Select the most appropriate answer.

How many of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



✓ (a) $[\text{Cp}_2\text{ZrMe}(\text{THF})]\text{BPh}_4$

(b) $\text{WCl}_6/\text{AlMe}_3$ ✓

✓ (c) $\text{Br}_2(\text{RO})_2\text{Mo}=\text{CHPh}$

(d) $(\text{RO})_3\text{Mo}\equiv\text{CCMe}_3$ ✓

(e) two of them

(f) three of them

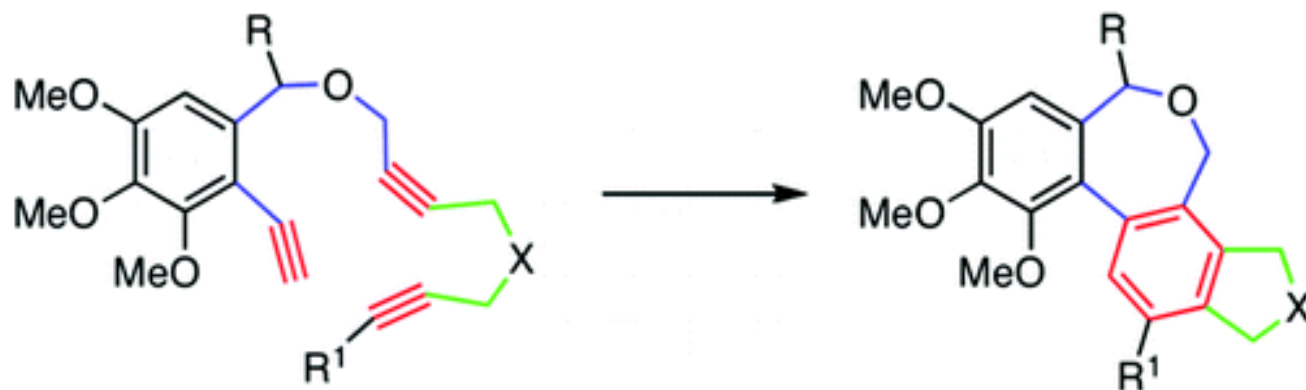
(g) all of them

(h) none of them



Answer: (g)

Q15. How many of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



✓ (a) $\text{Cp}^*\text{RuCl}(\text{COD})$

(b) $\text{RhCl}(\text{PPh}_3)_3$ ✓

✓ (c) PdCl_2

(d) $\text{CpCo}(\text{COD})$ ✓

(e) two of them

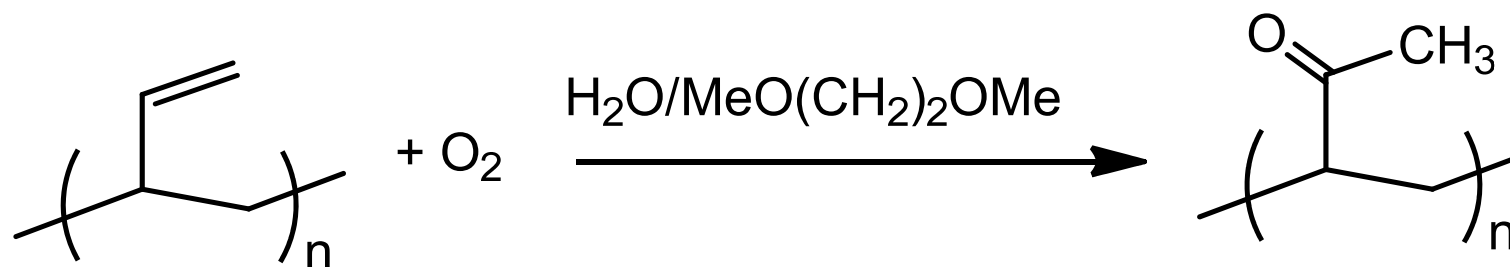
(f) three of them

(g) all of them

(h) none of them

Answer: (g)

Q9. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



Wacker oxidation

✓

(a) $\text{PdCl}_2/\text{CuCl}$

(b) $(\text{Cy}_3\text{P})_2\text{Cl}_2\text{Ru}=\text{CHPh}$ ✗

✓

(c) $\text{PdCl}_2/\text{CuCl}_2$

(d) $\text{TiCl}_4/\text{AlEt}_3$ ✗

✓

(e) two of them

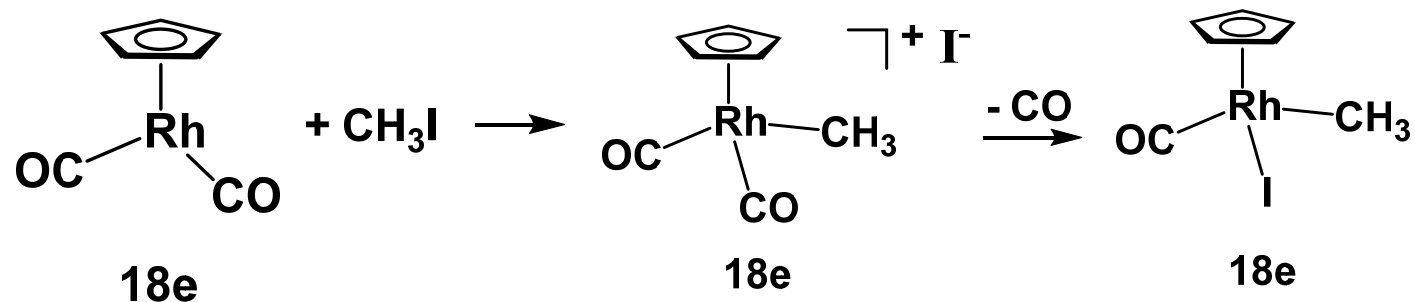
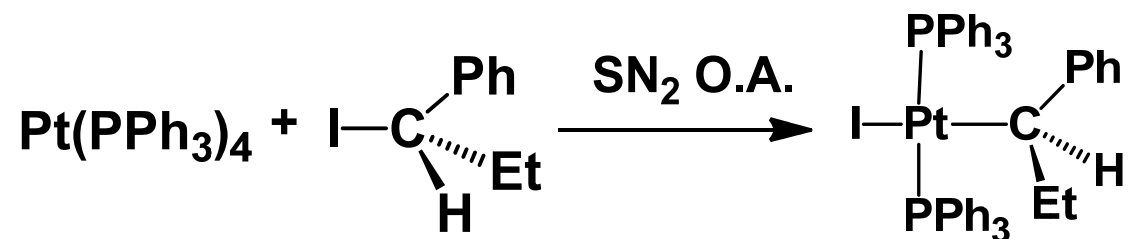
(f) three of them

(g) all of them

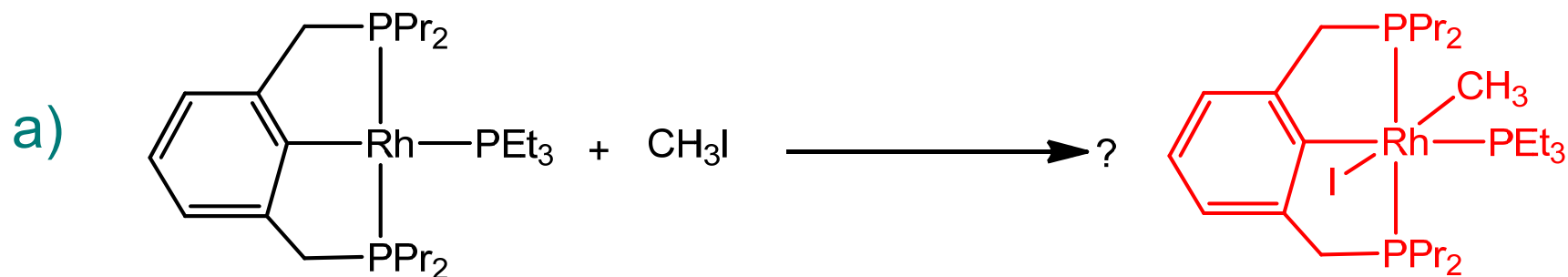
(h) none of them

Answer: (e)

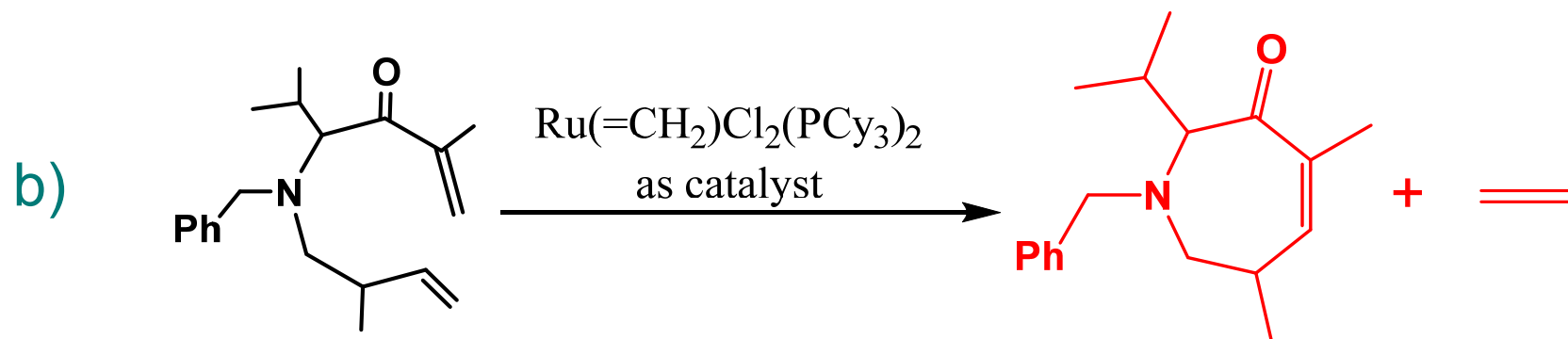
3. Predict the product for reactions



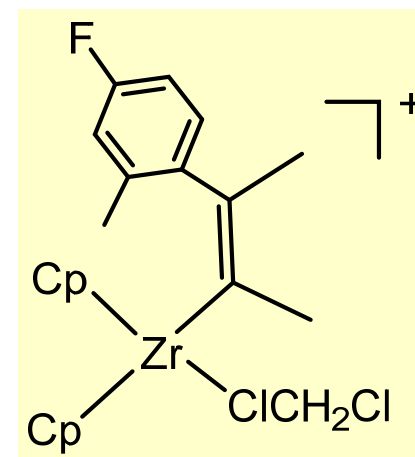
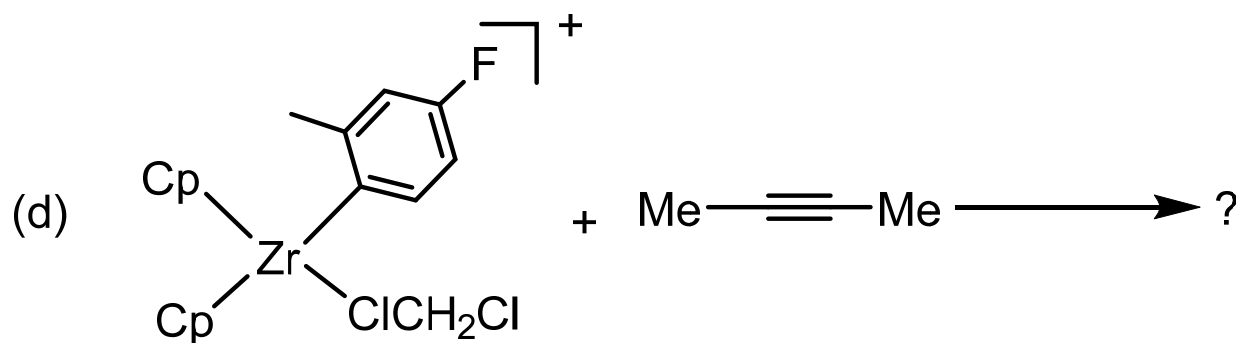
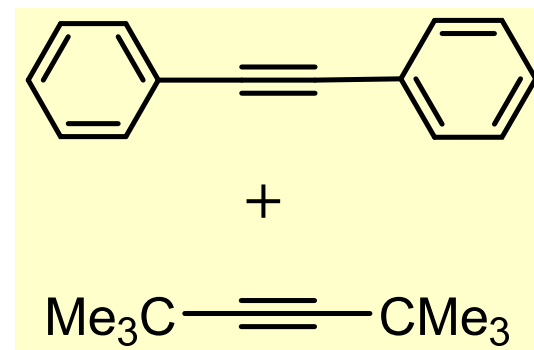
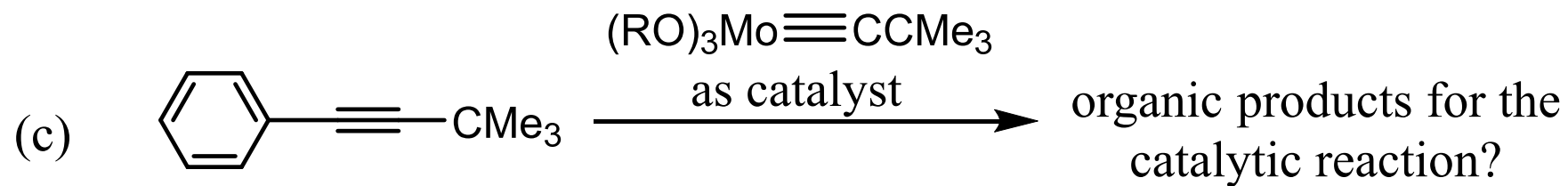
3. Predict the products.



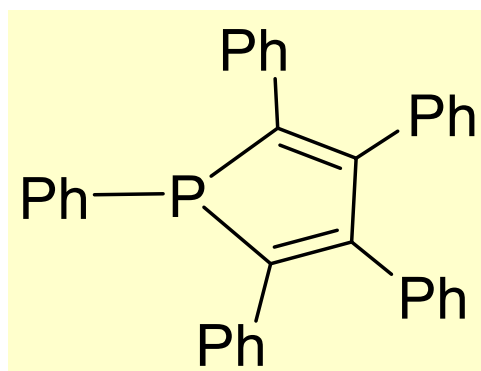
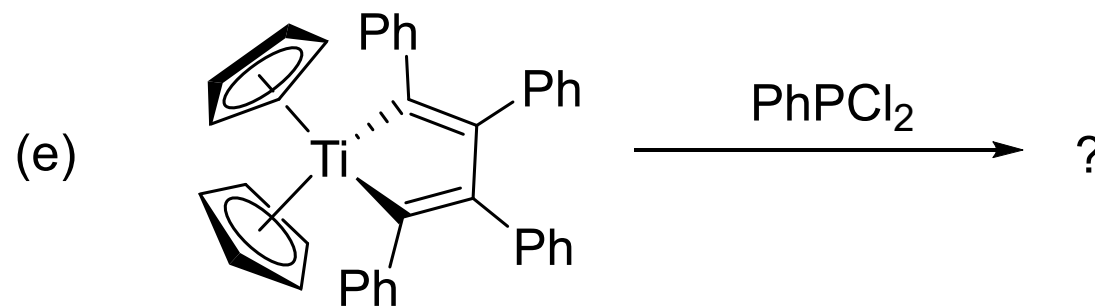
(Ref. *Organometallics*, **2009**, 28, 523)



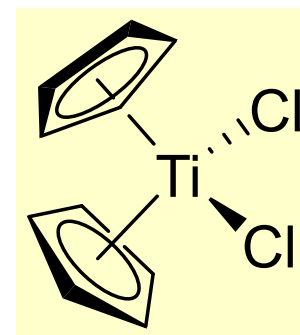
3. Predict the products.



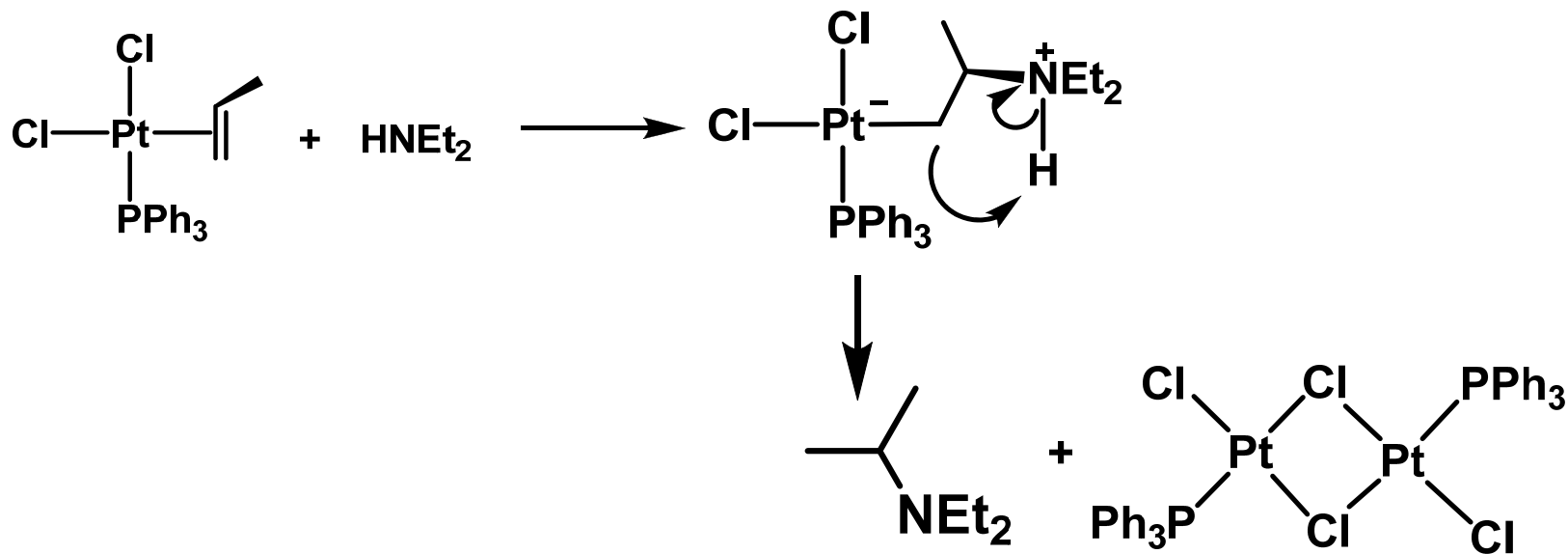
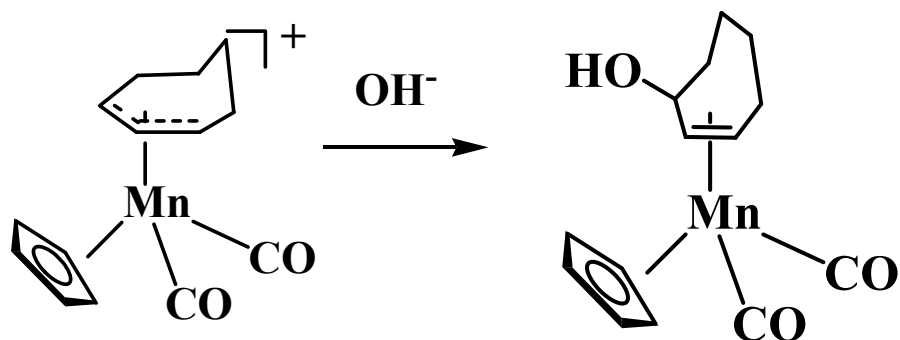
3. Predict the products.



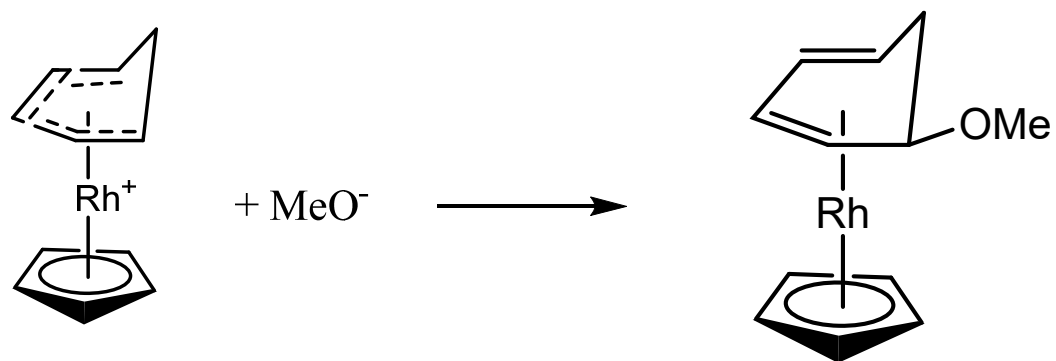
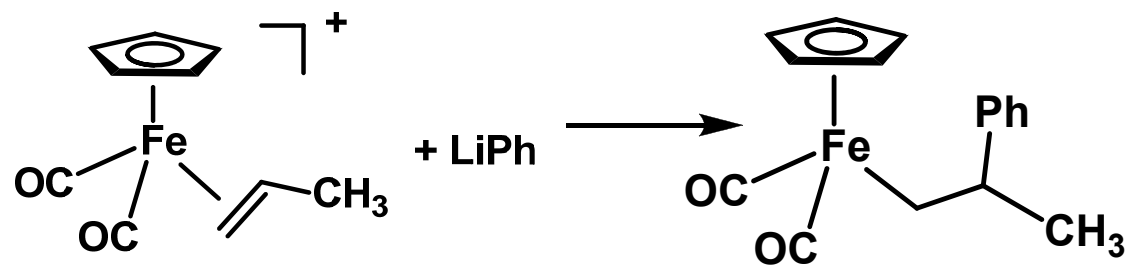
+



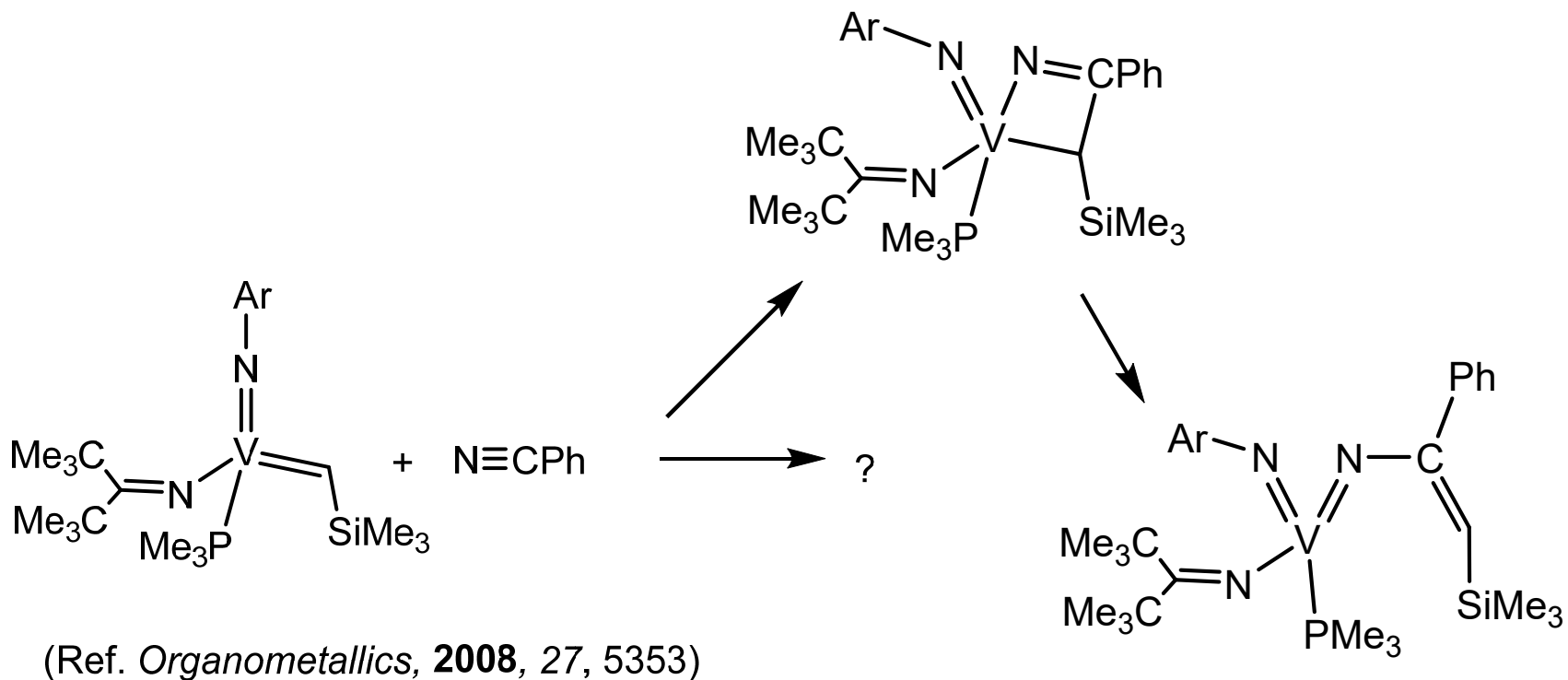
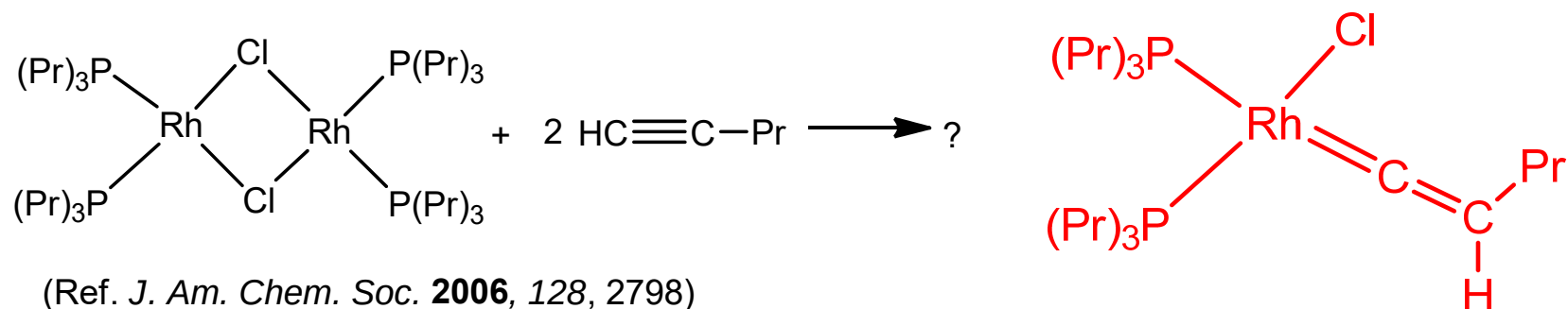
Give the products of the following reactions.



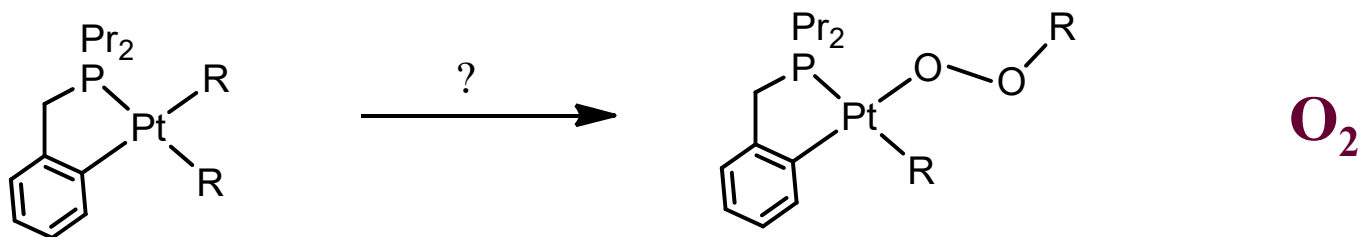
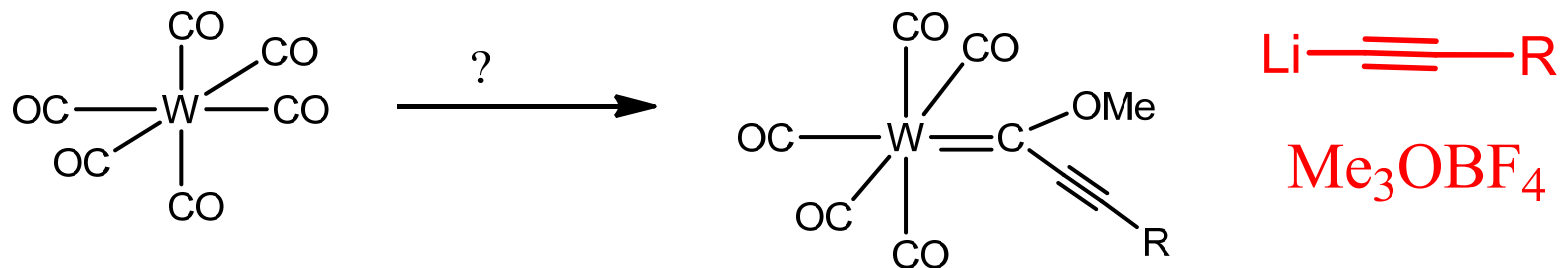
Give the products of the following reactions.



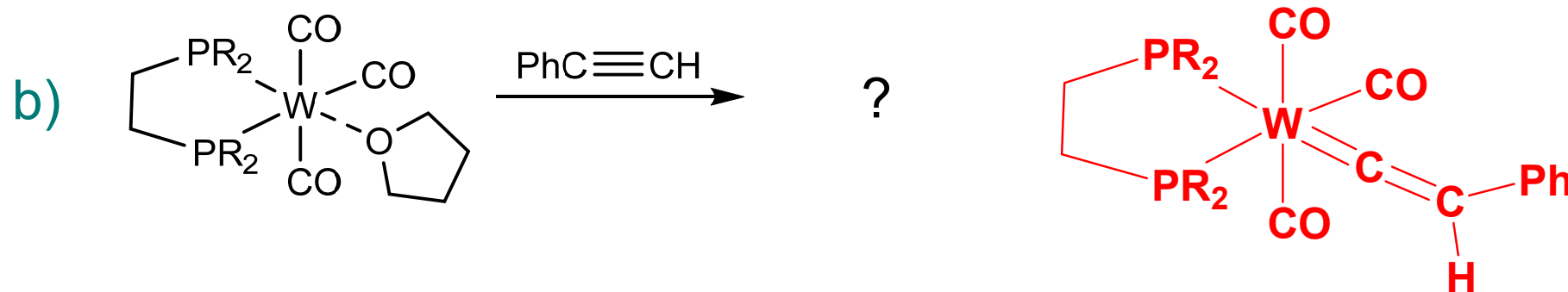
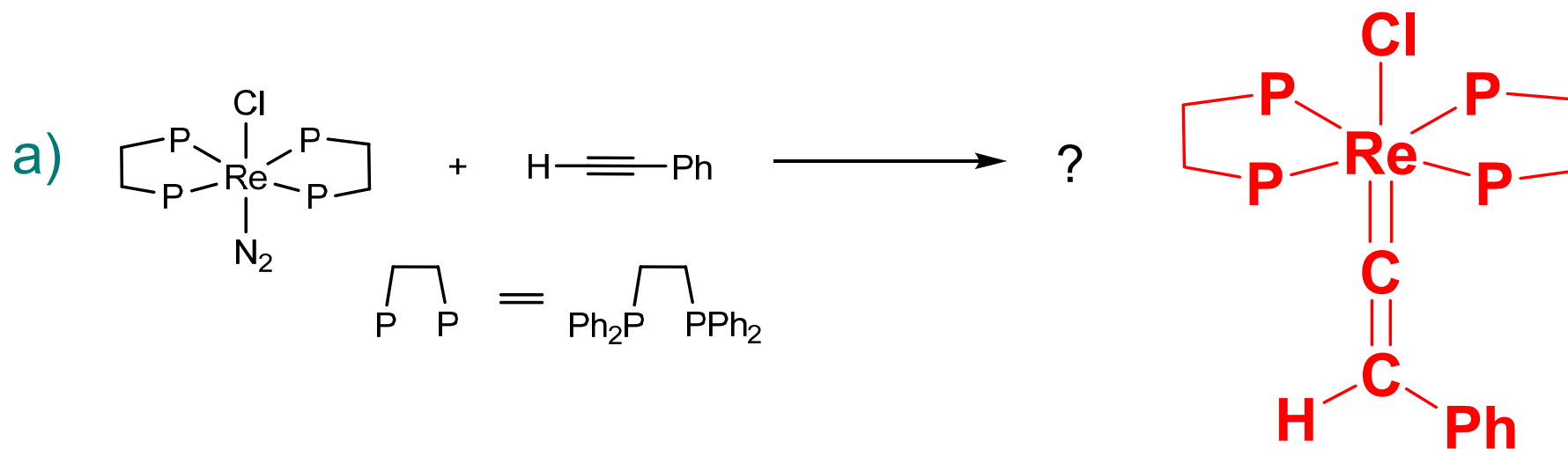
Predict the metal-containing products of the following reactions.

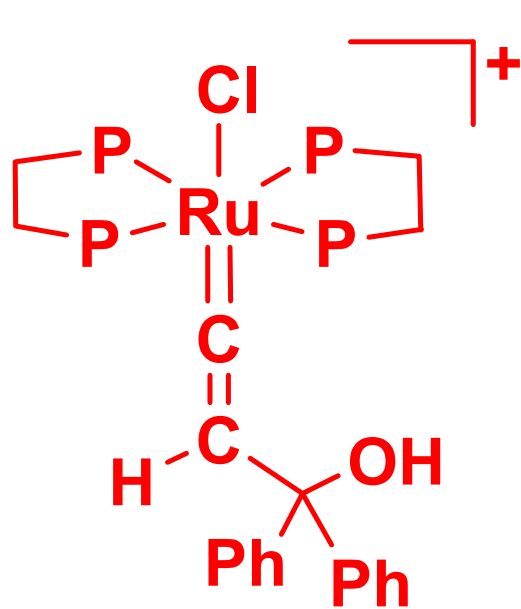
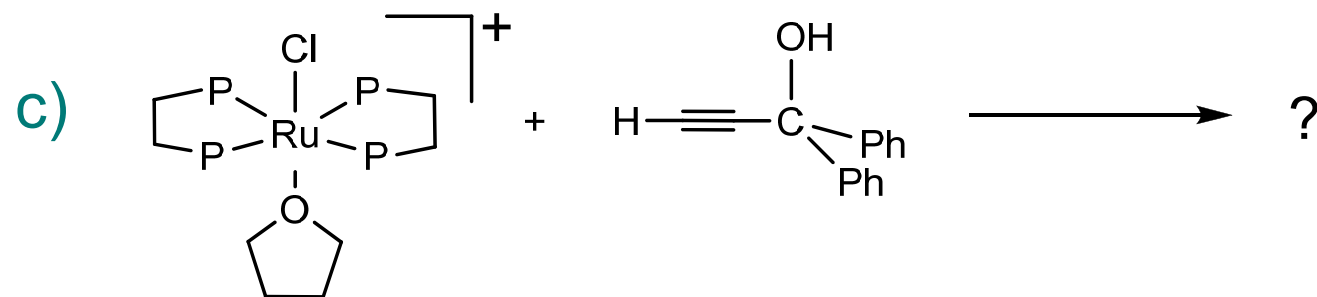


Provide appropriate reagents for each of the following transformations.

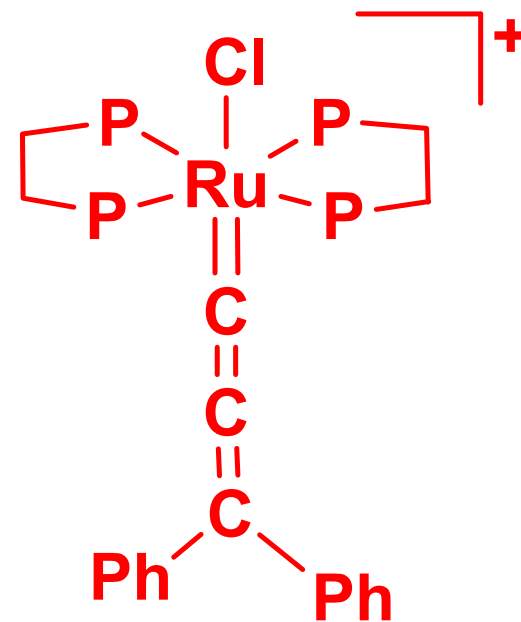


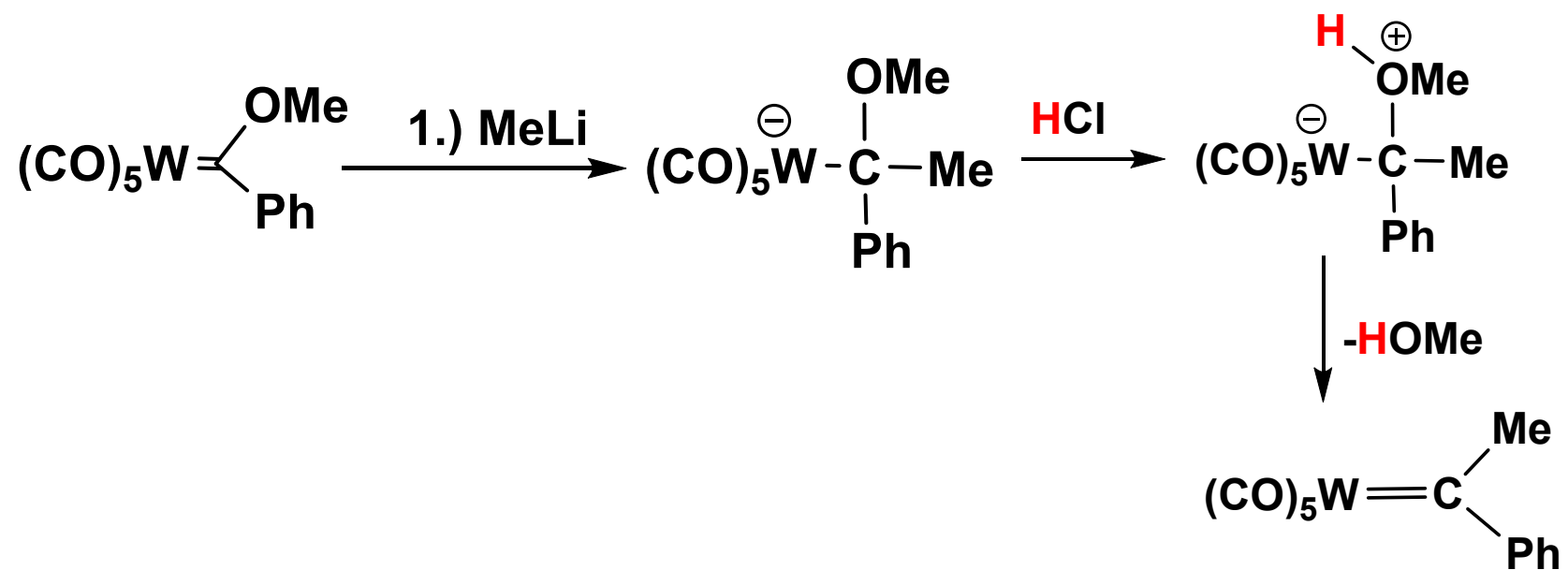
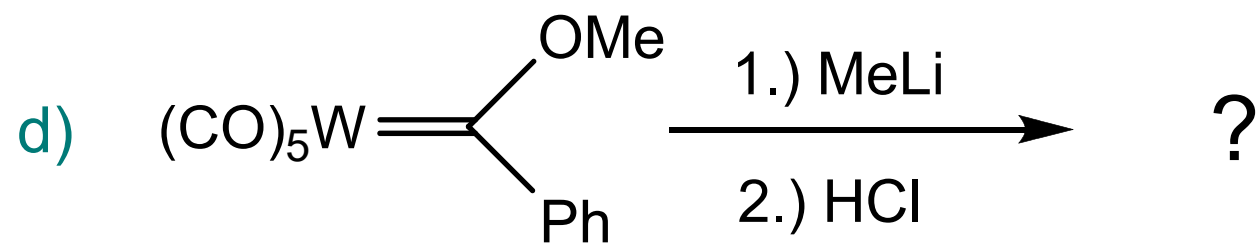
1. Predict the products.





or

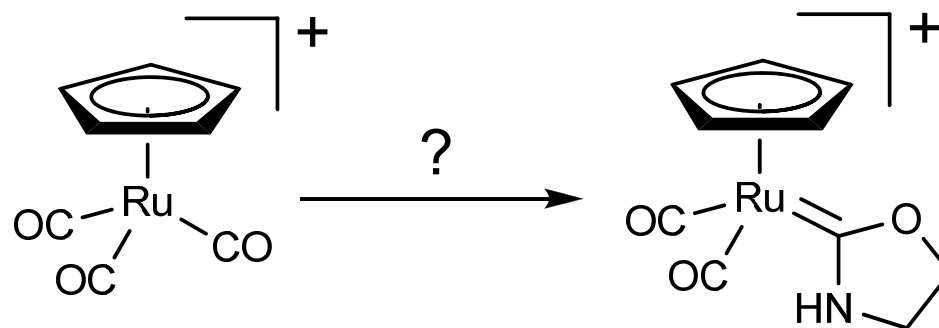






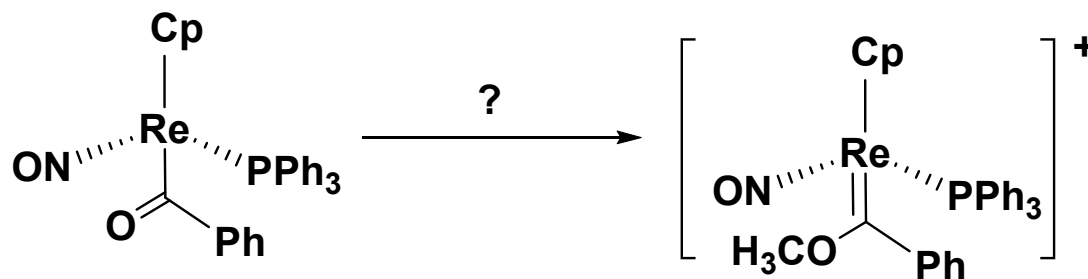
2. Suggest reagents

a)



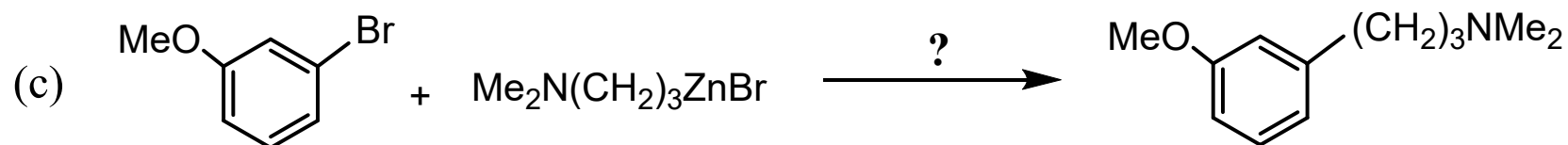
$\text{NH}_2\text{CH}_2\text{CH}_2\text{Br}/\text{base}$

b)



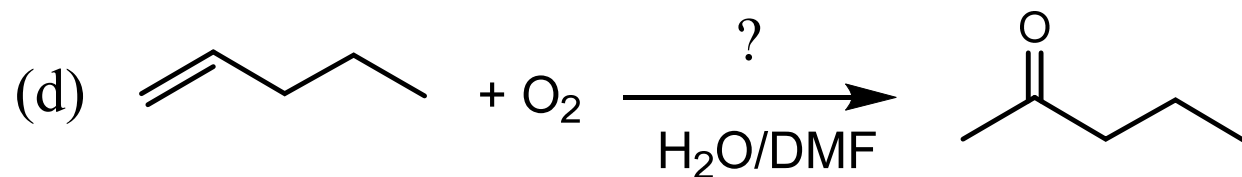
Me^+ , e.g. Me_3OBF_4 , $\text{CF}_3\text{SO}_3\text{CH}_3$

4. Suggest reagents or catalyst for the following reactions.



Cross Coupling

$\text{NiCl}_2(\text{PPh}_3)_2$ or $\text{PdCl}_2(\text{PPh}_3)_2$, $\text{Pd}(\text{PR}_3)_4$

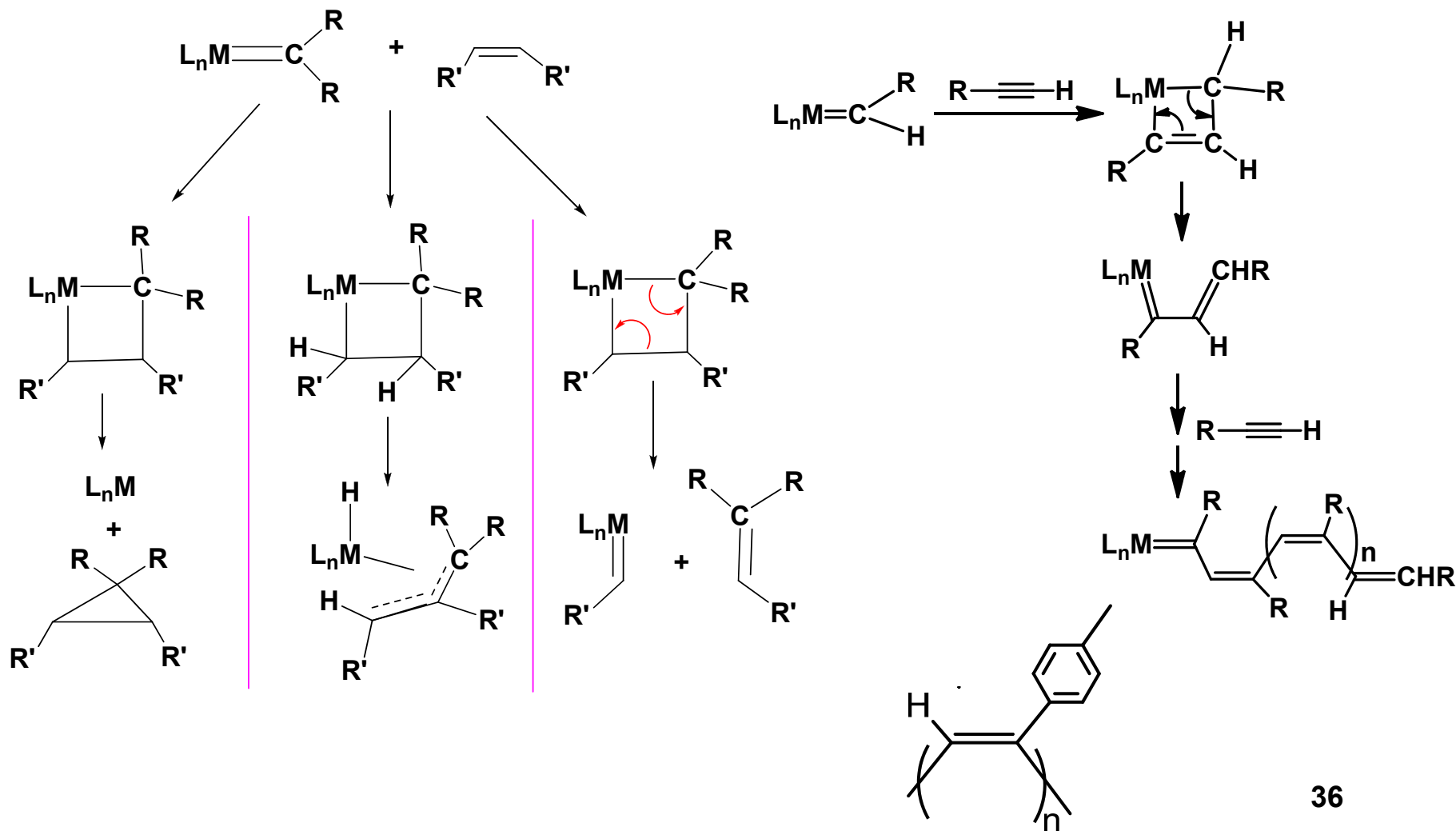


Wacker oxidation

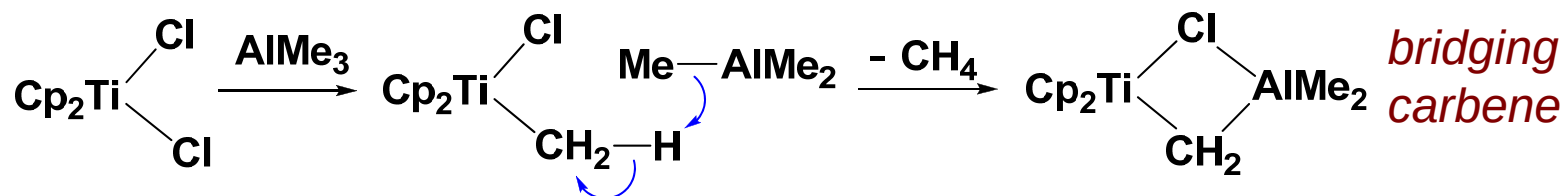
$\text{PdCl}_2/\text{CuCl}_2$ or $\text{PdCl}_2/\text{CuCl}$

Reactions of carbene complexes with olefins and acetylenes

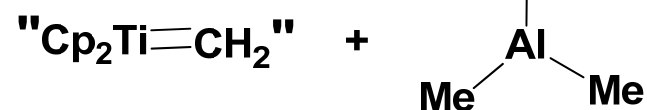
There are at least three possible products for the reactions, depending on metal complexes and olefins used.



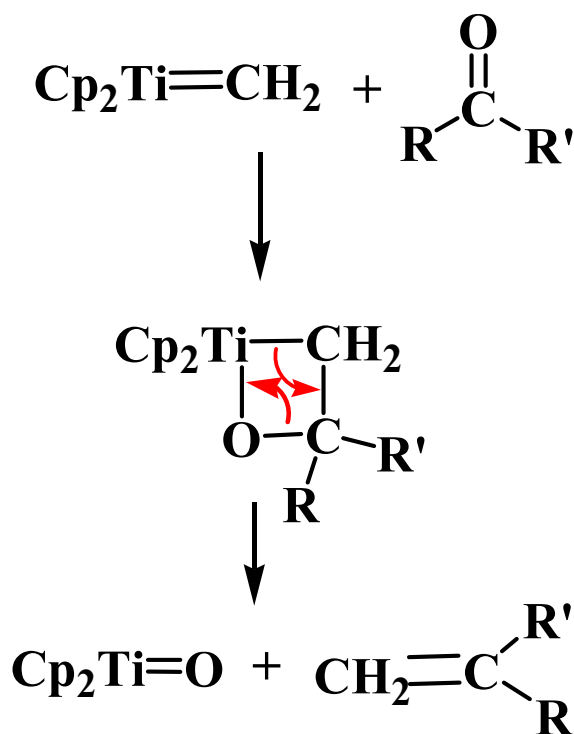
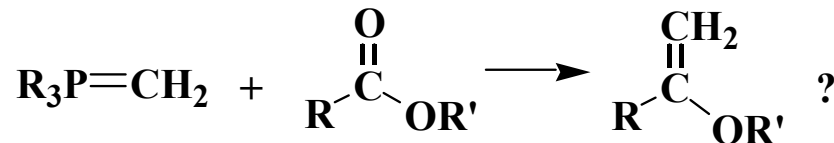
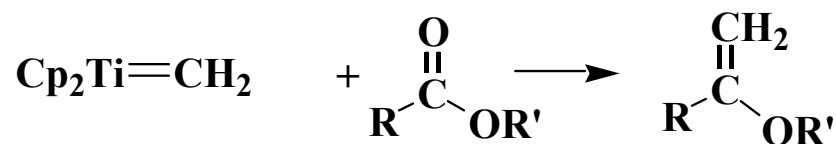
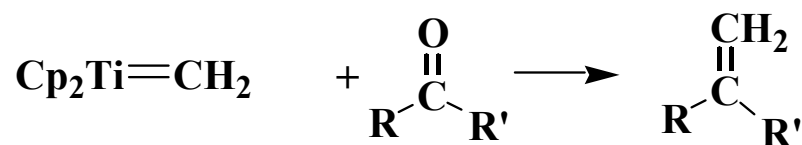
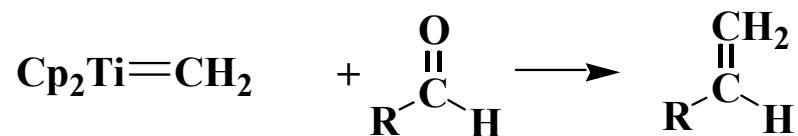
Tebbe's reagent:



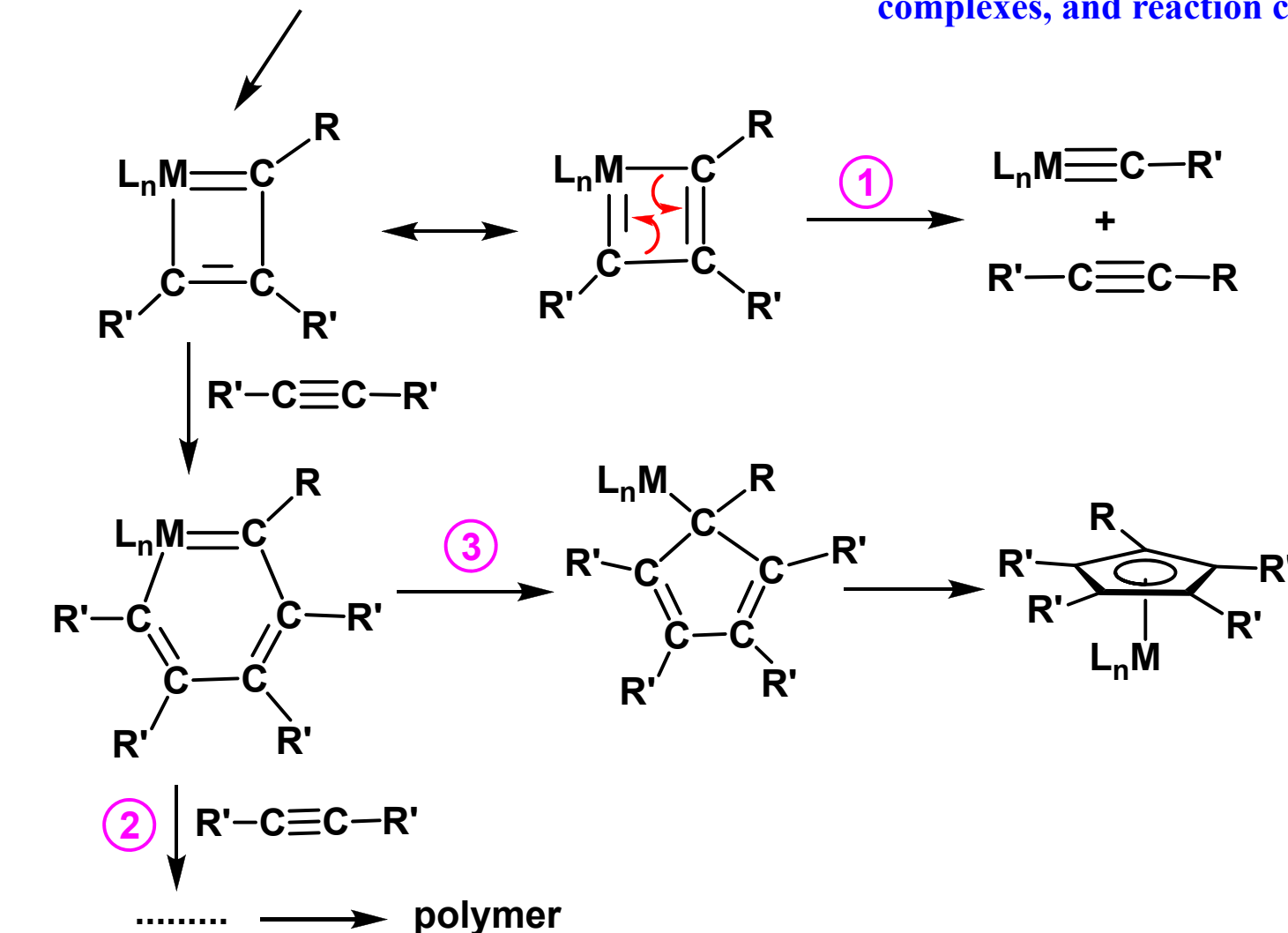
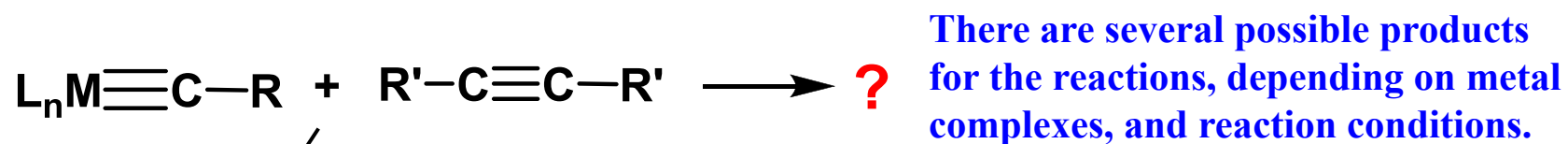
Lose Me_2AlCl first to give mononuclear carbene



Even give Wittig-type product with esters

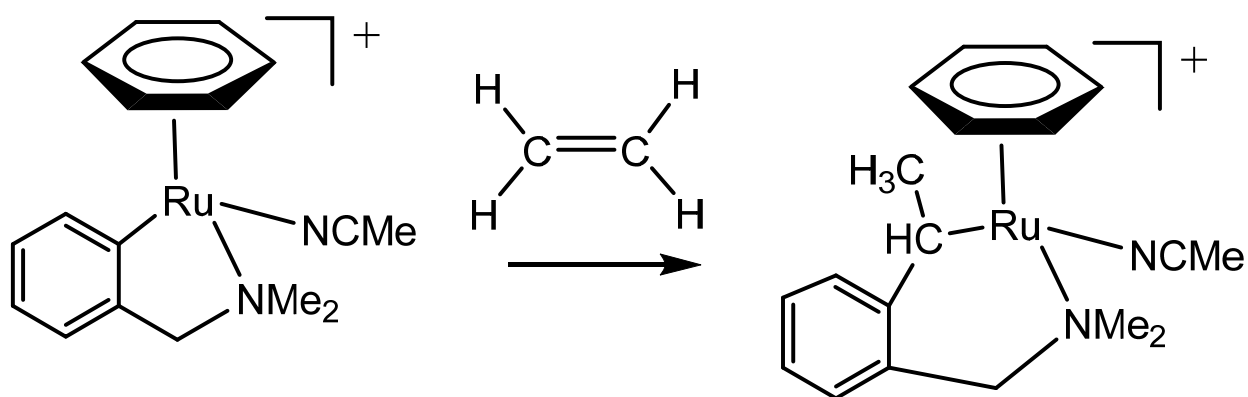


Review: Reactions of carbyne complexes with acetylenes

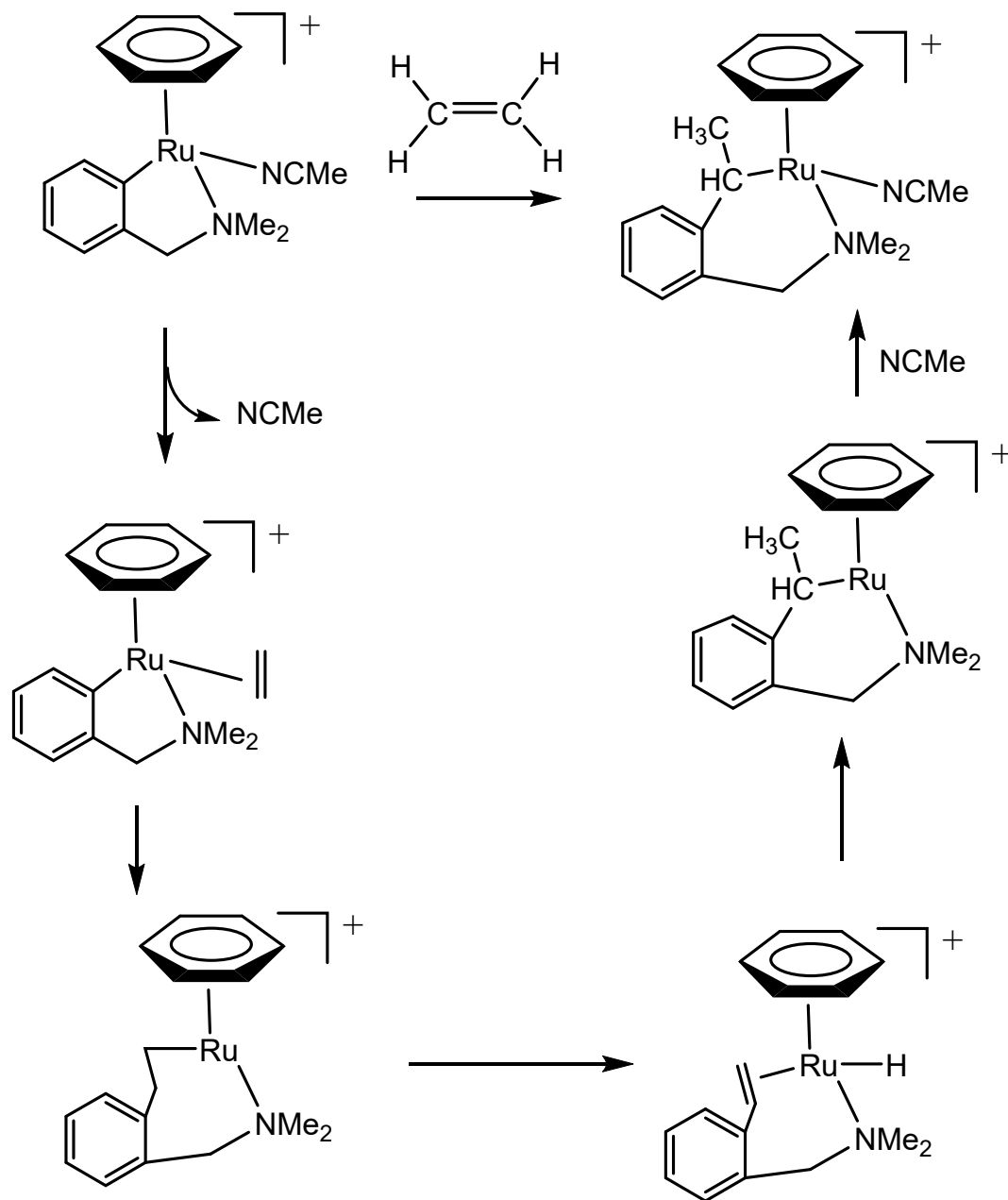


5. Mechanism for stoichiometric reactions

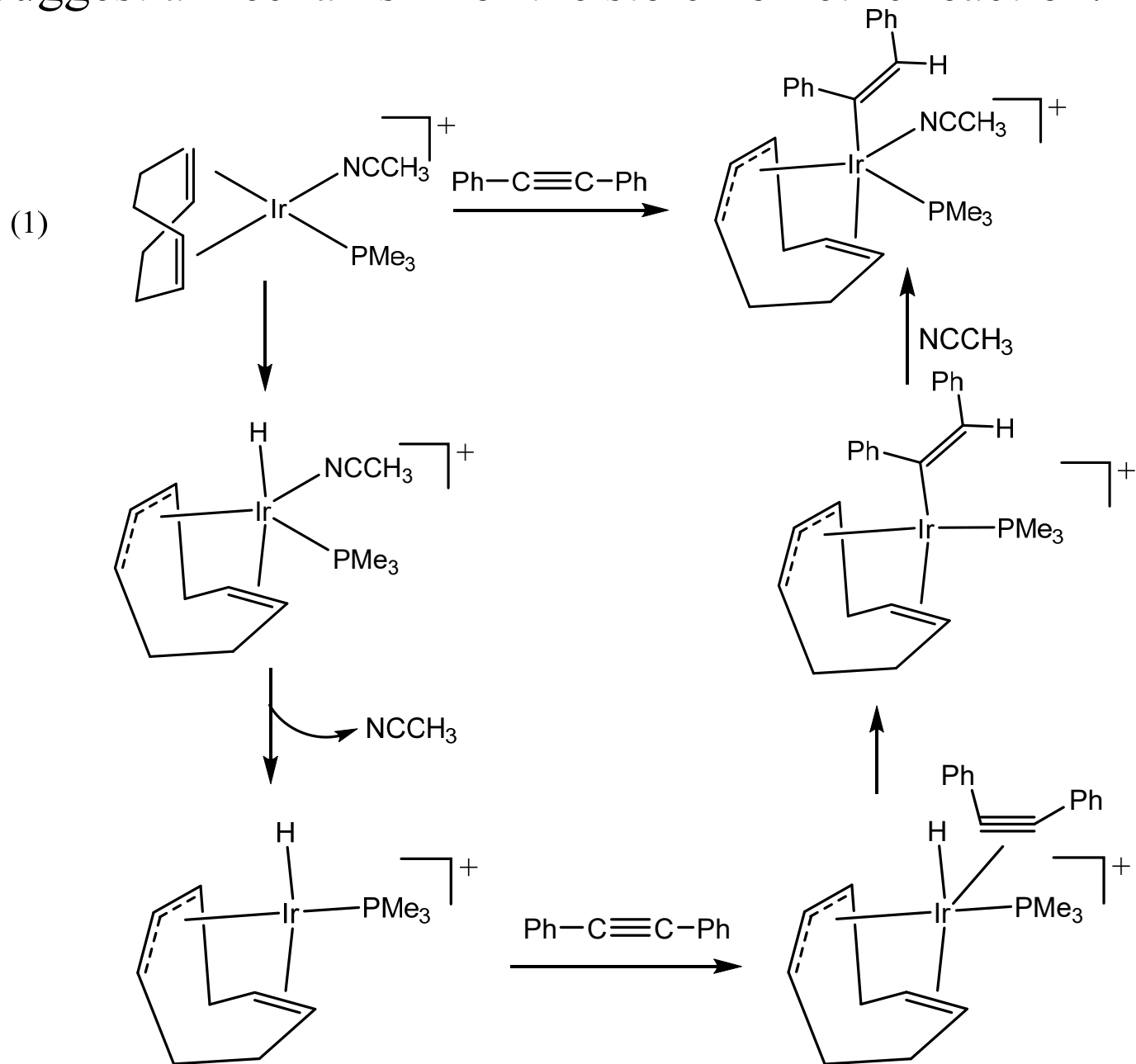
1. Propose a mechanism for the following reactions.



1. Propose a mechanism for the following reactions.

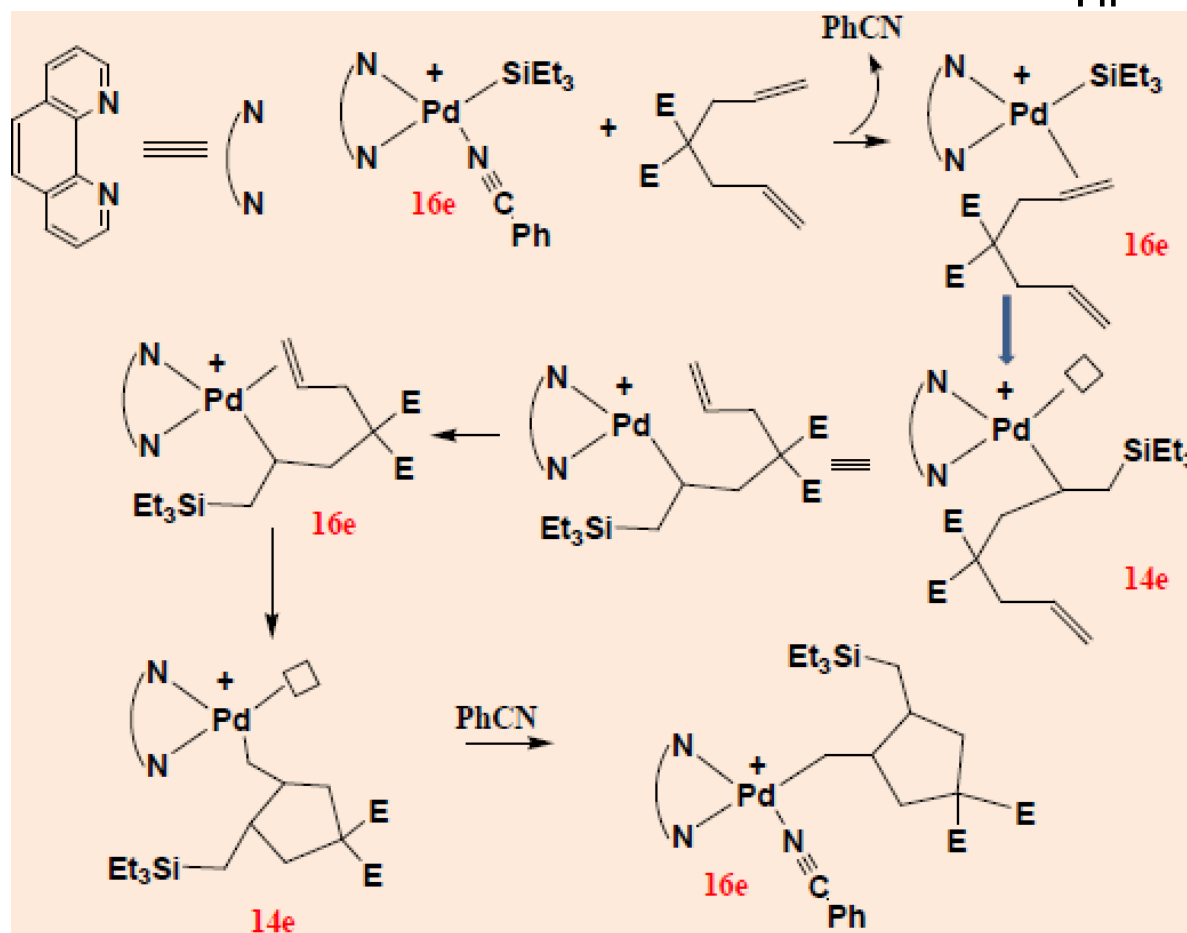
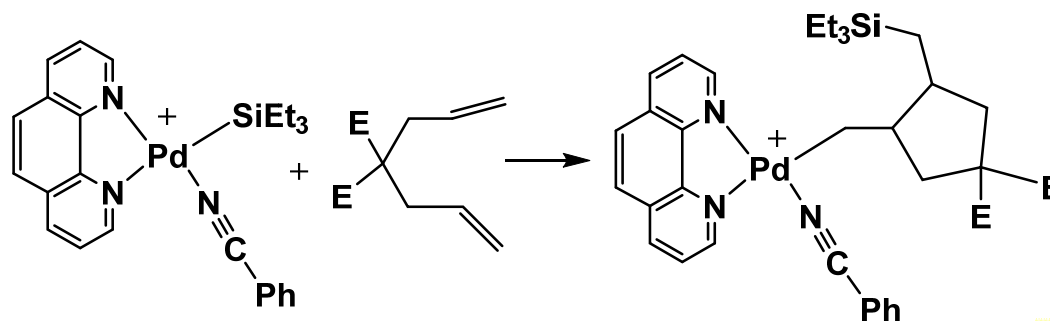


2. Suggest a mechanism for the stoichiometric reaction.



Review questions

2. Suggest a reasonable mechanism for the following reaction.

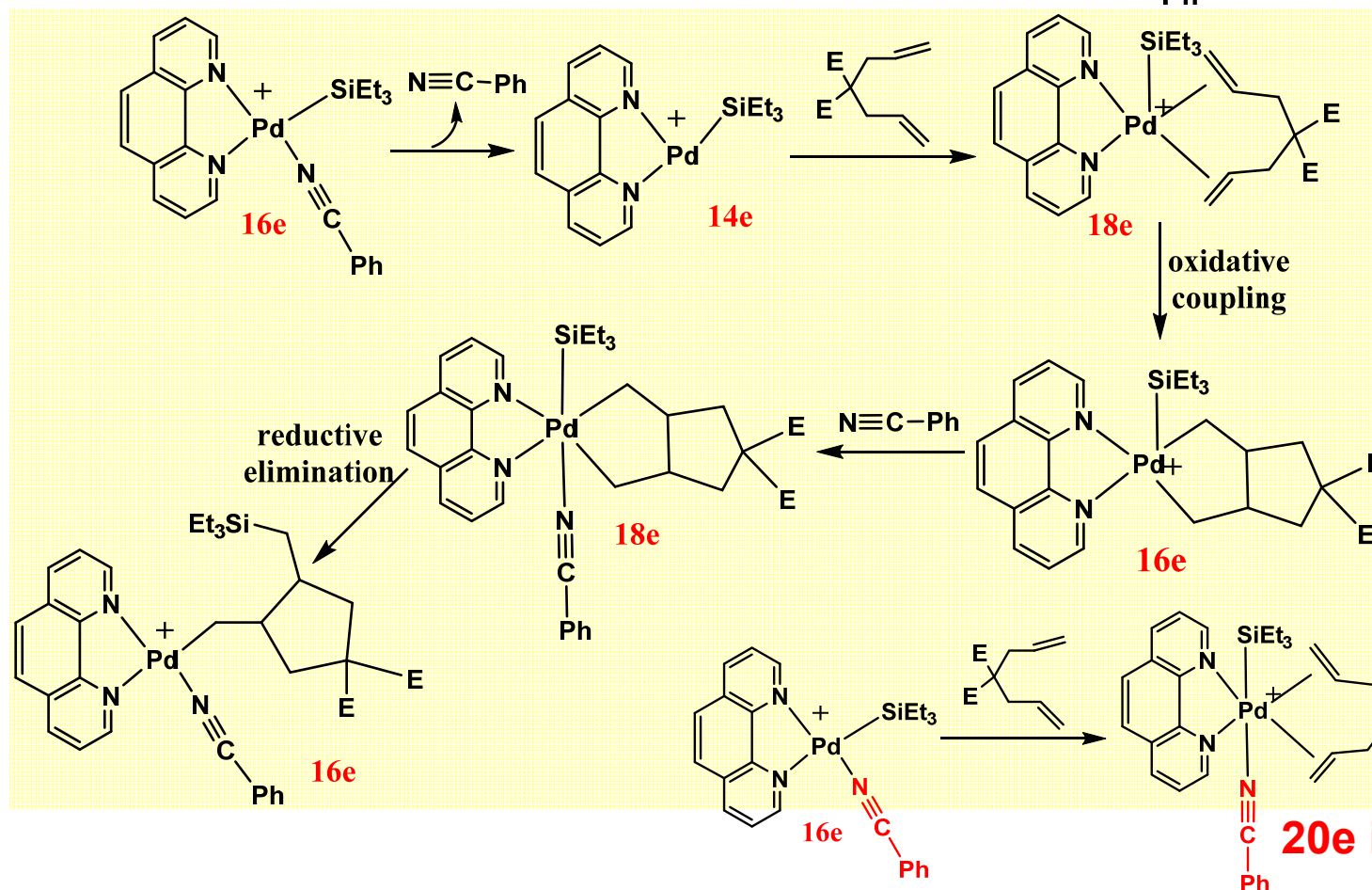
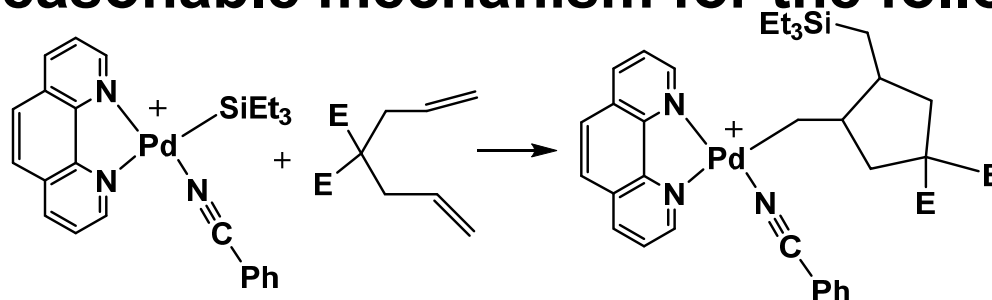


Prefer Pd(II) 16e square-planar intermediate.



Review questions

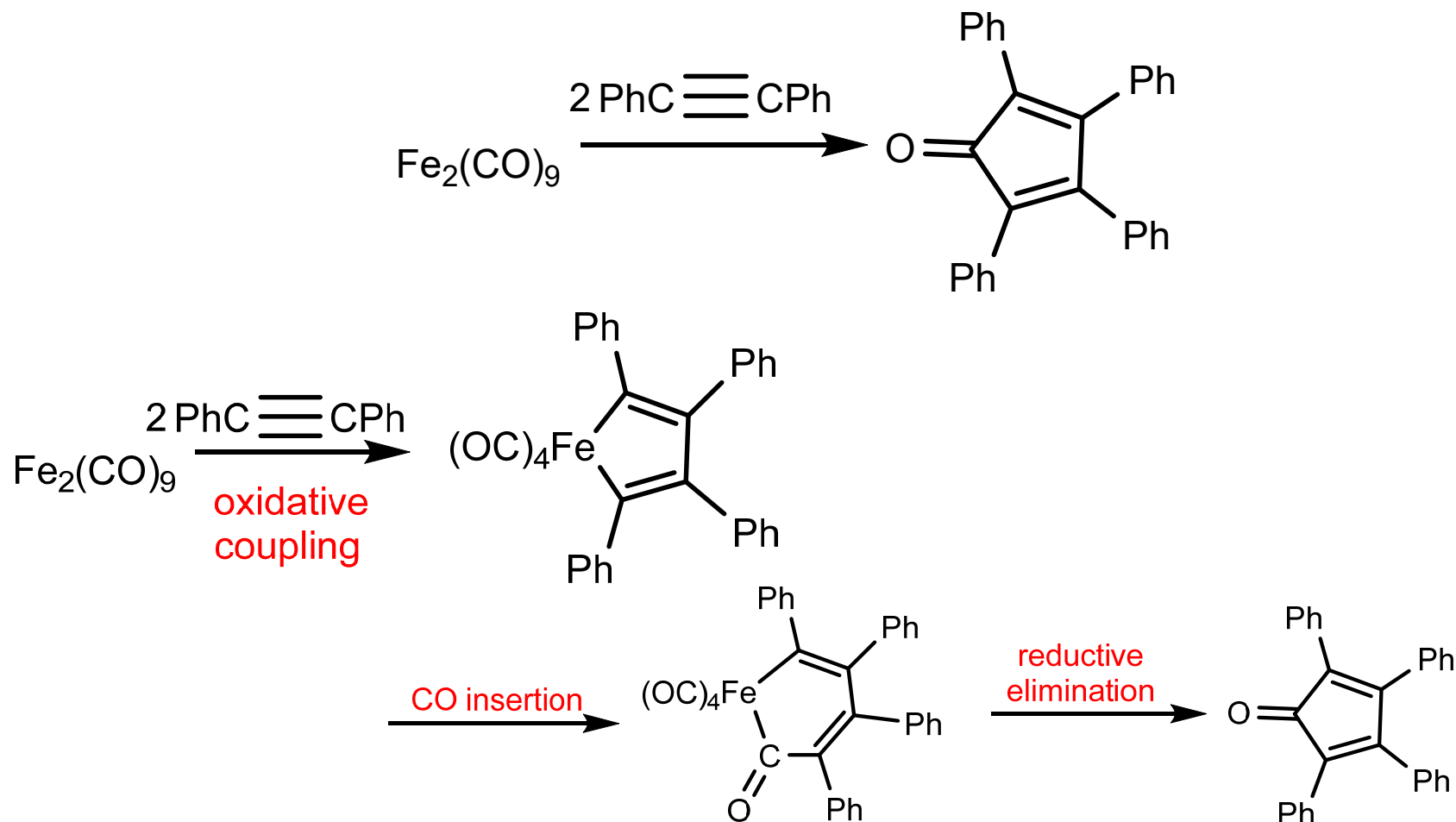
2. Suggest a reasonable mechanism for the following reaction.



5-coordinate
18e Pd(II)
intermediate
may also have
a certain
degree of
rationality. ✓

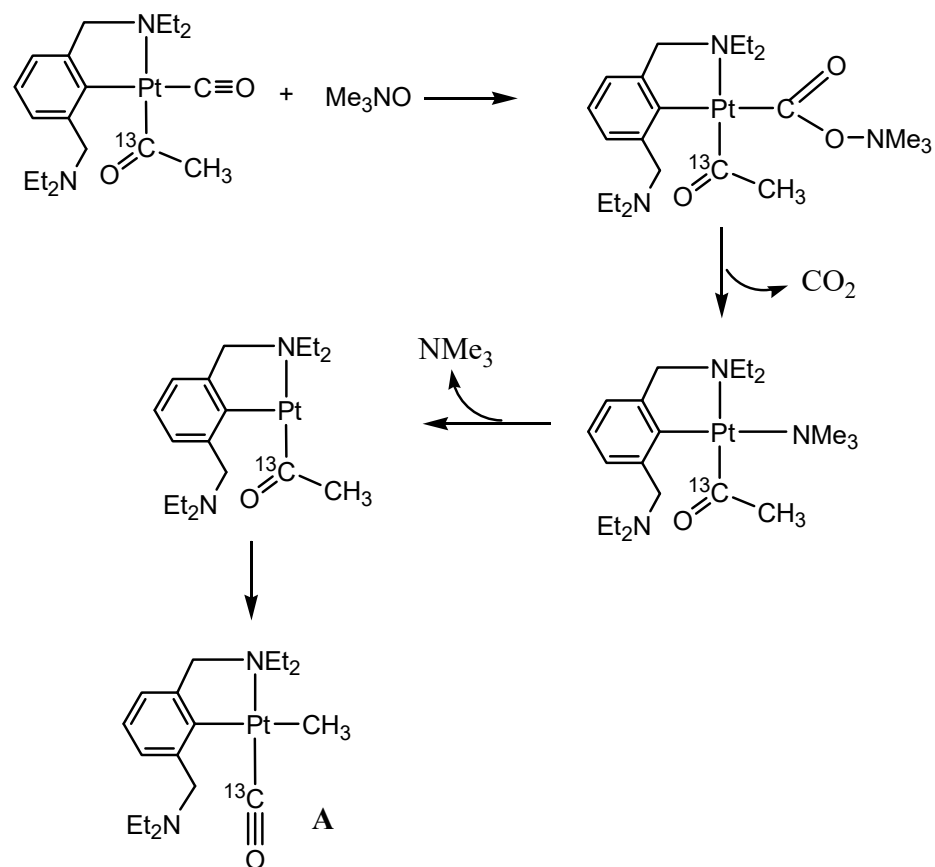
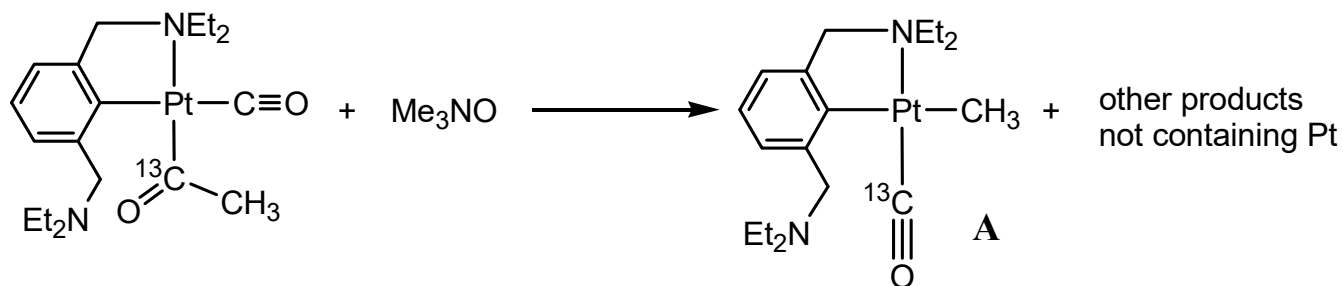
20e intermediate ✗

Propose a mechanism for the formation of 2,3,4,5,-tetraphenylcyclopentadienone from following reaction.

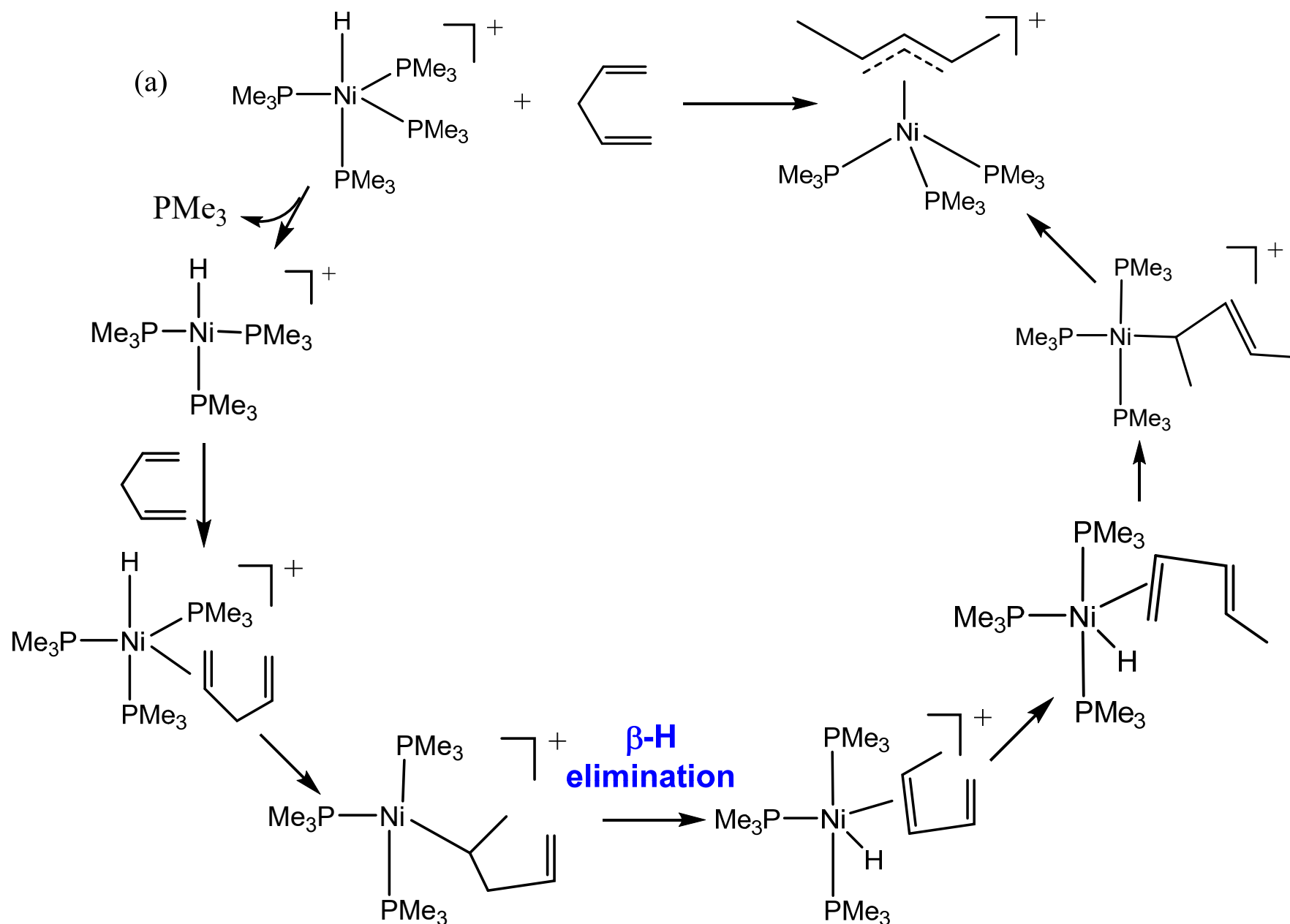


Oxidative coupling of two alkynes to give the metallole, followed by CO insertion and reductive elimination.

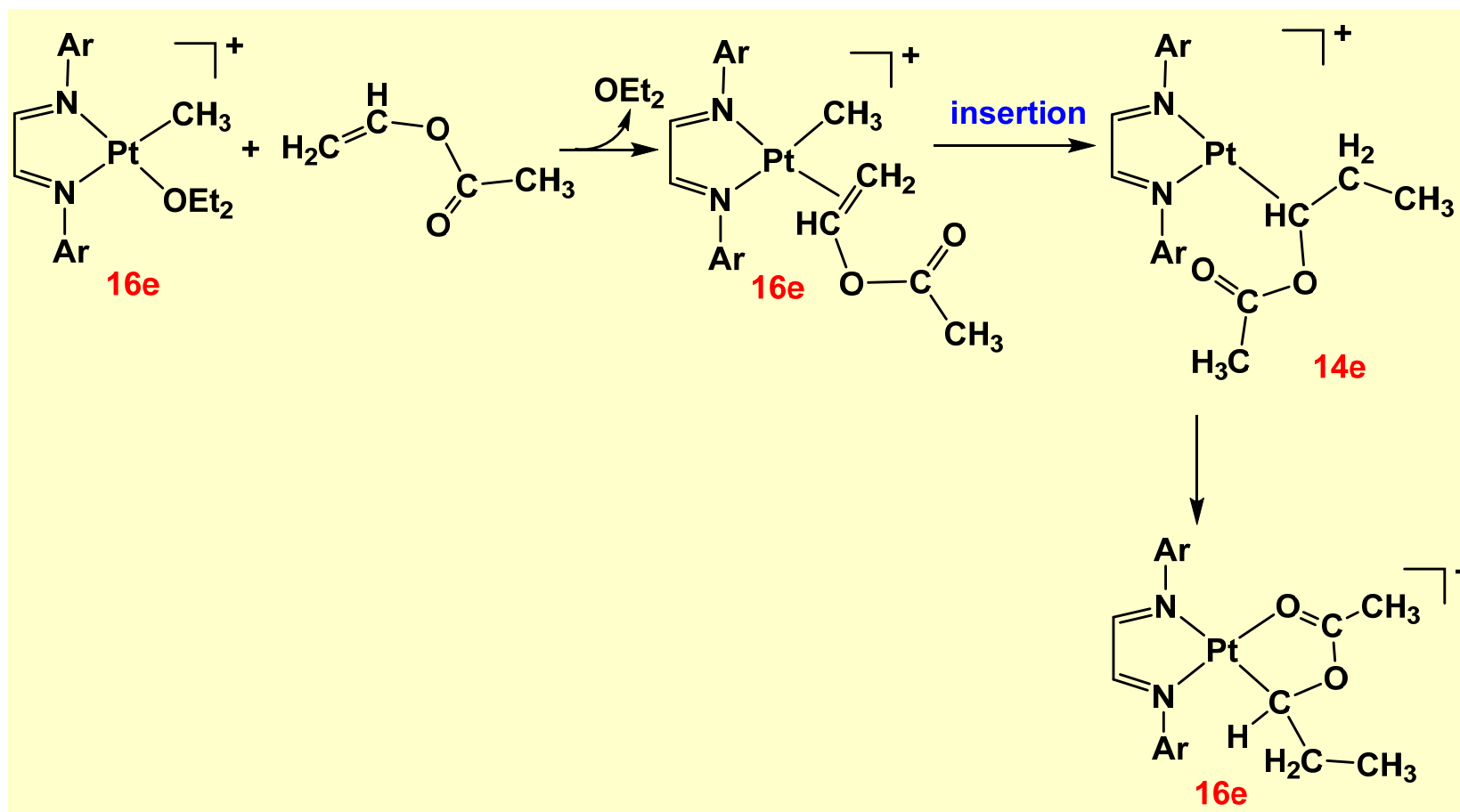
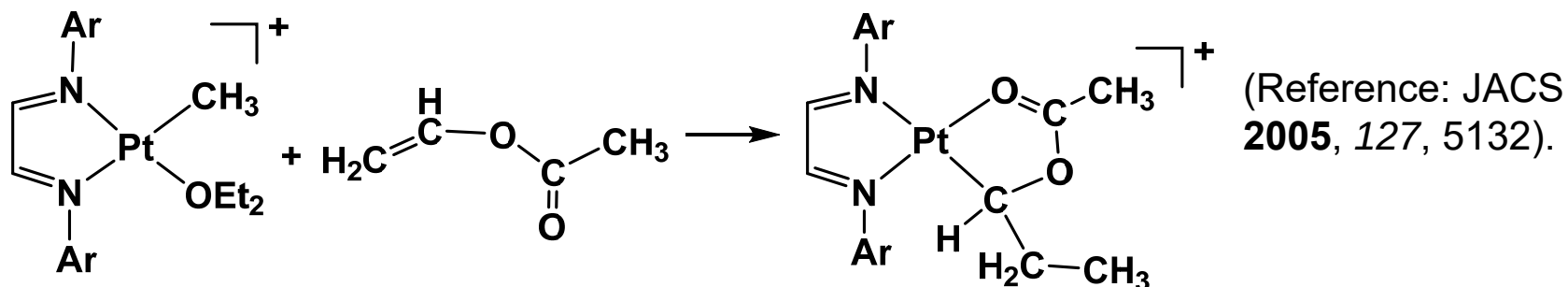
3. Propose a mechanism for the formation of product **A**. Be sure to indicate what other products are generated in the reaction.



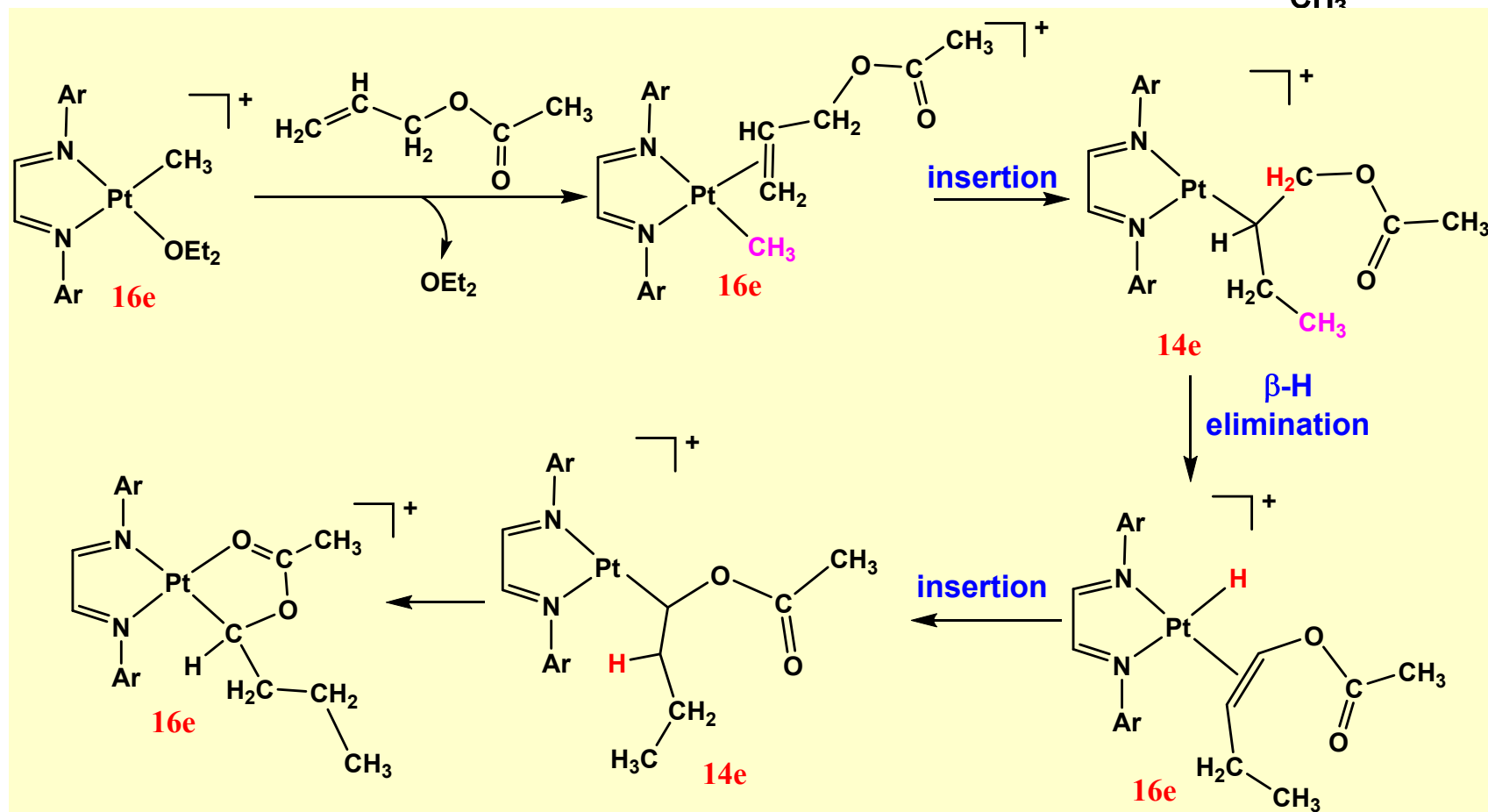
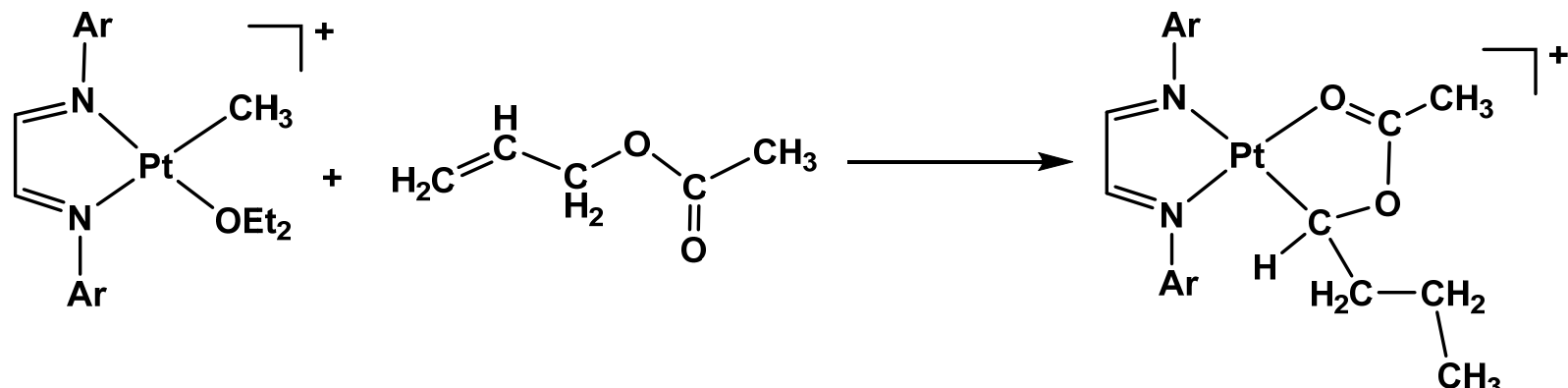
4. Propose a mechanism for the following reactions.



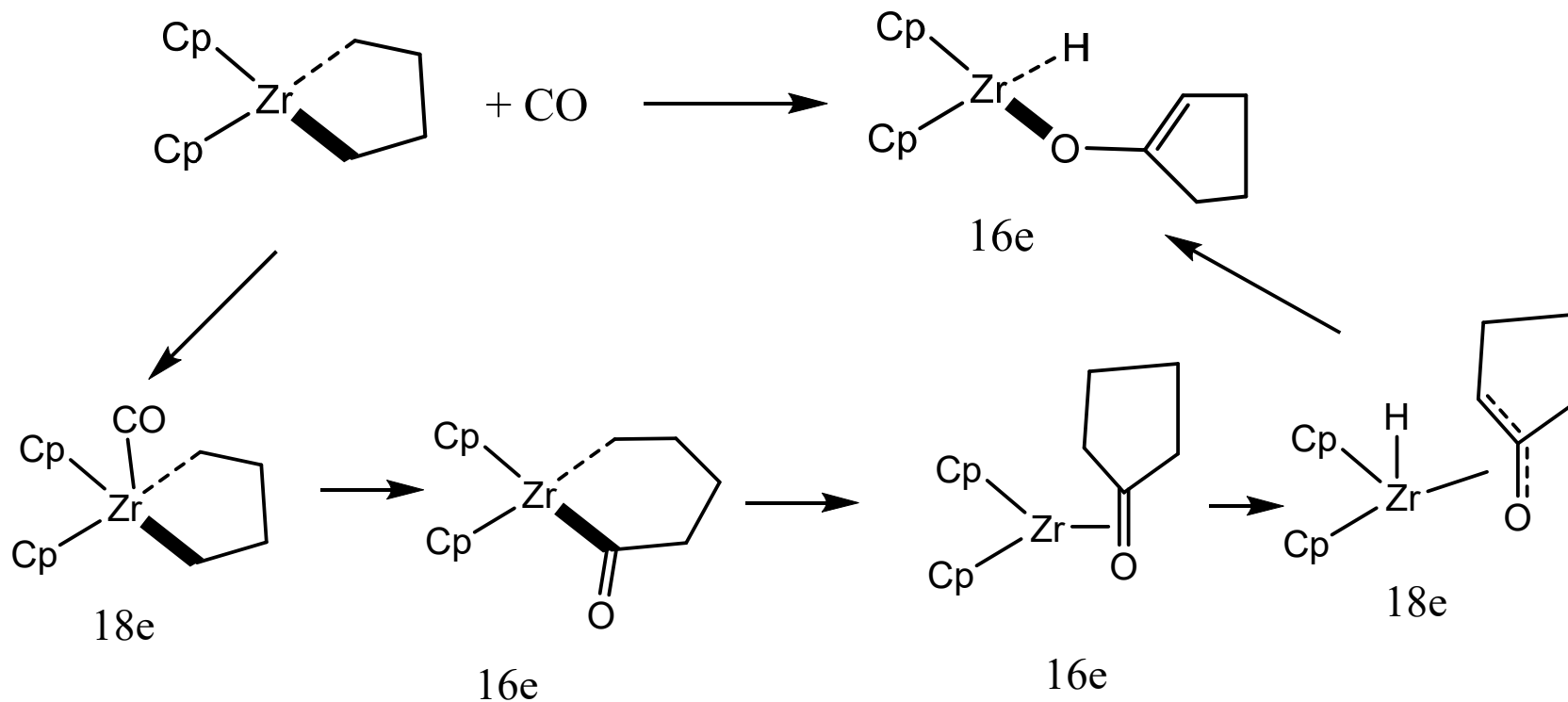
5. Suggest a reasonable mechanism for the following reaction.



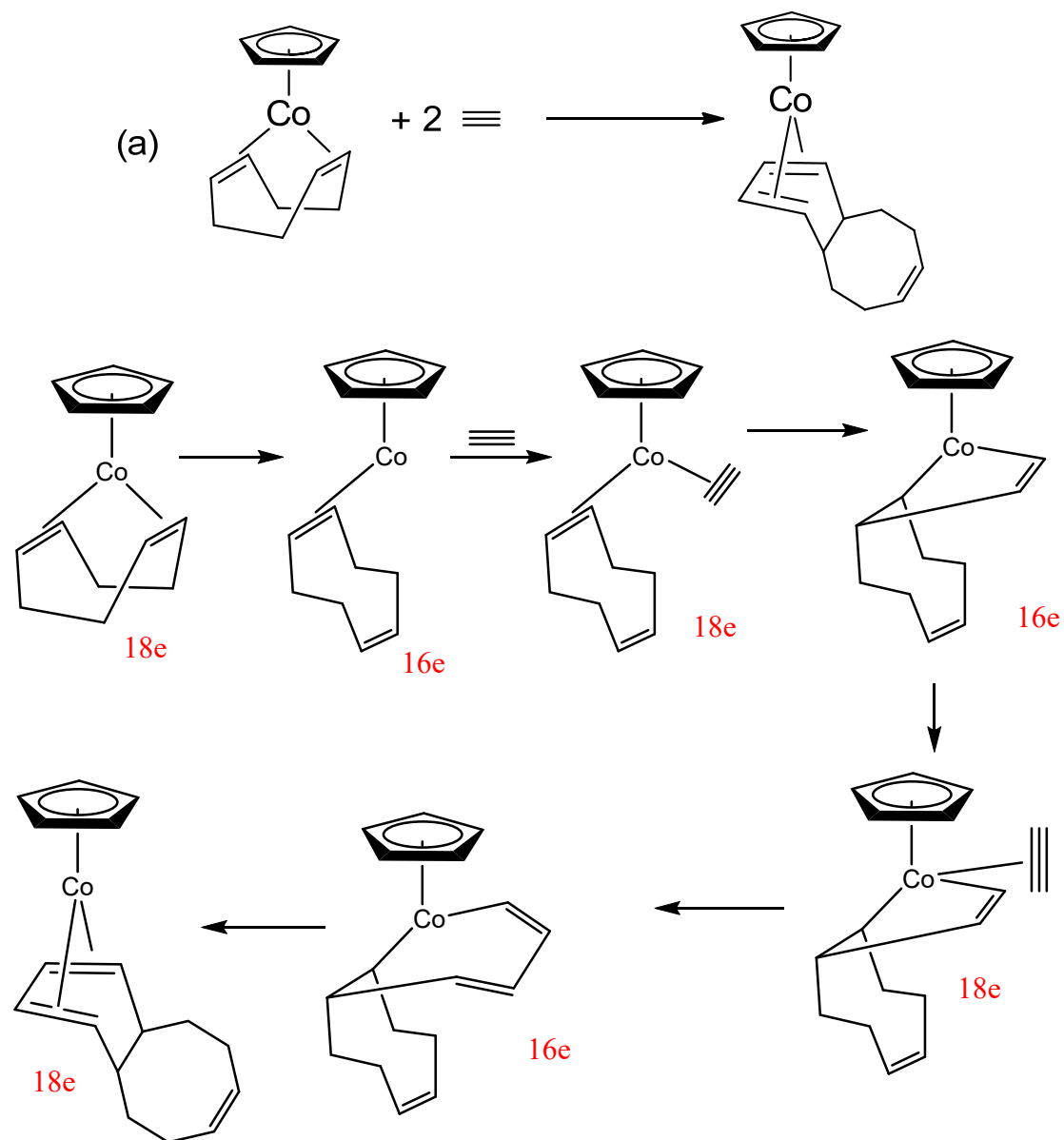
Suggest a reasonable mechanism for the following reaction



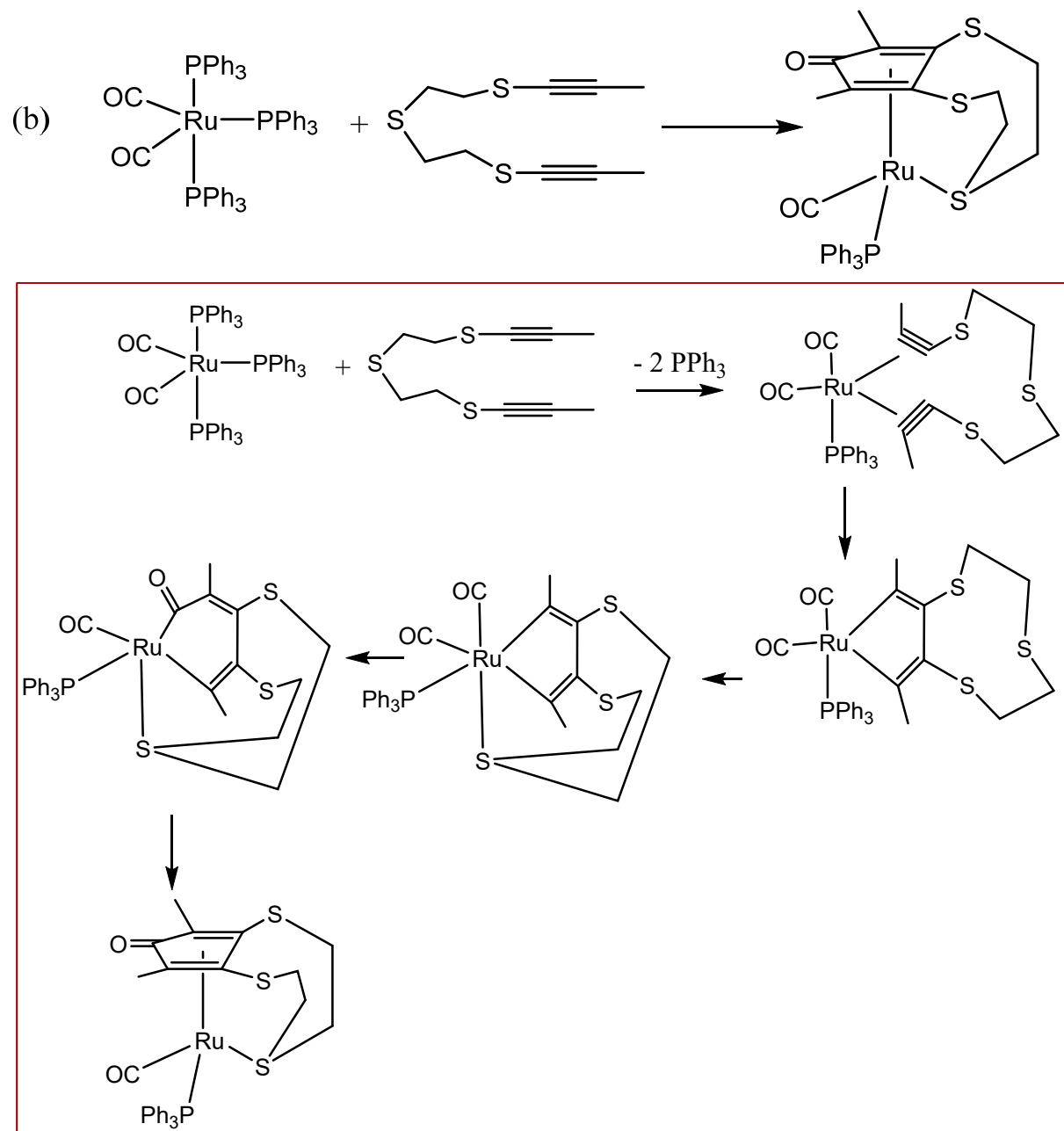
Suggest a reasonable mechanism for the following reaction



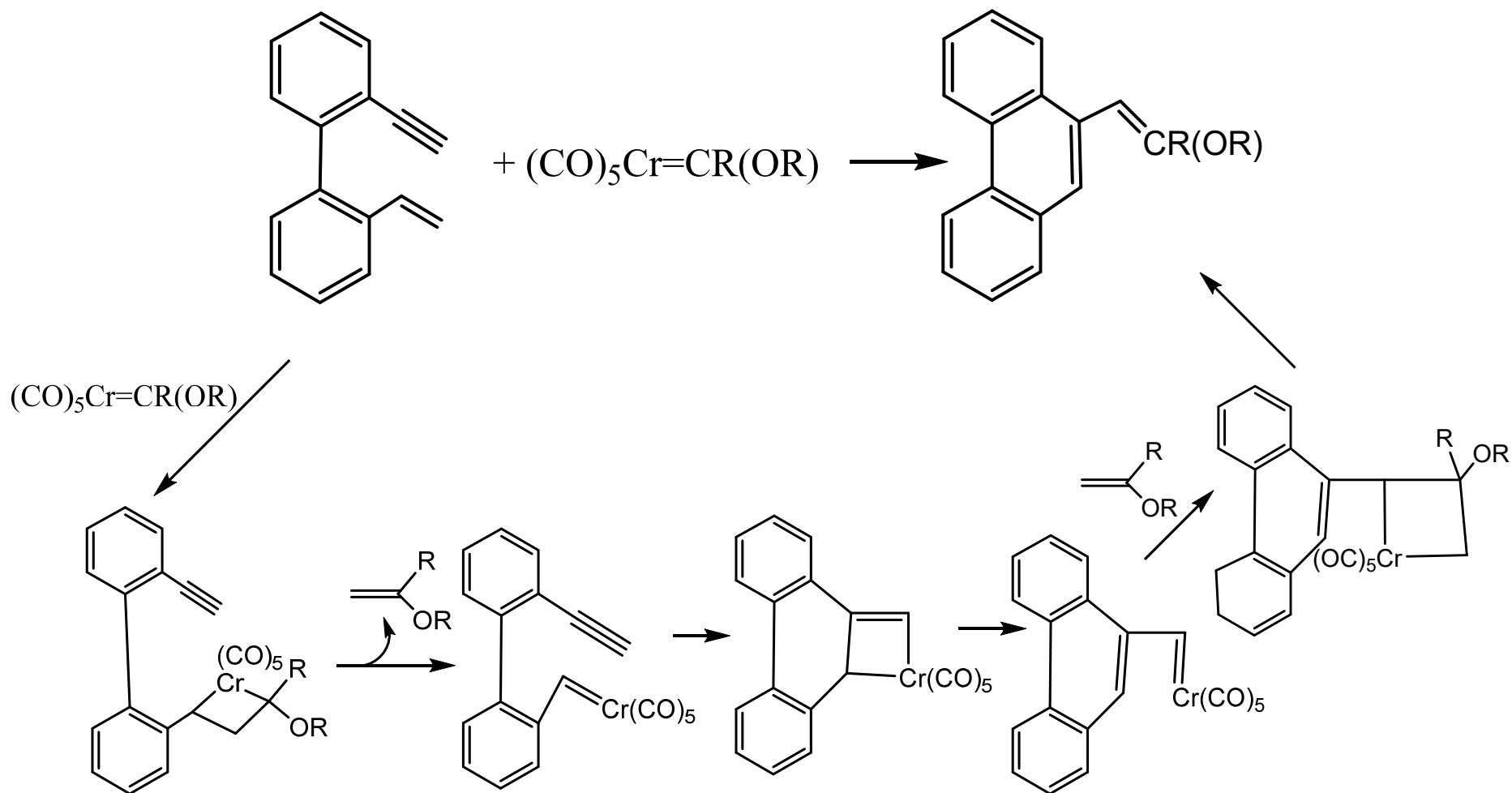
Propose a mechanism for the following reaction.

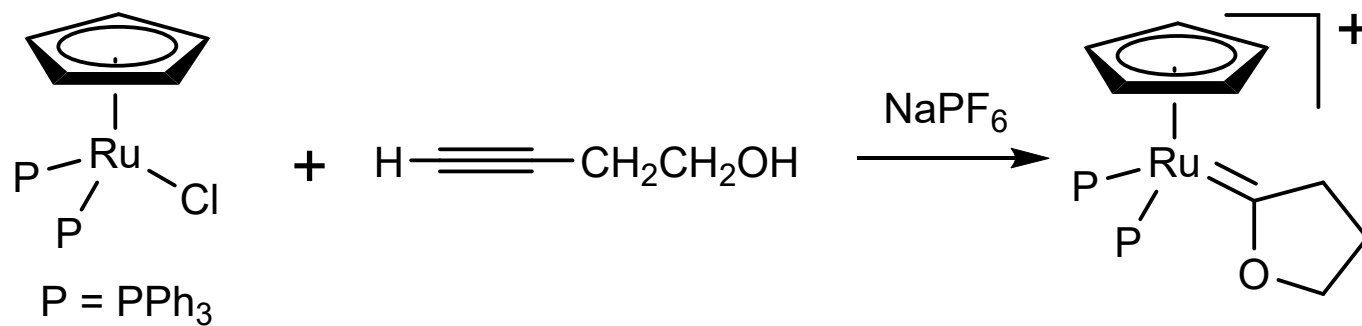


Propose a mechanism for the following reactions.

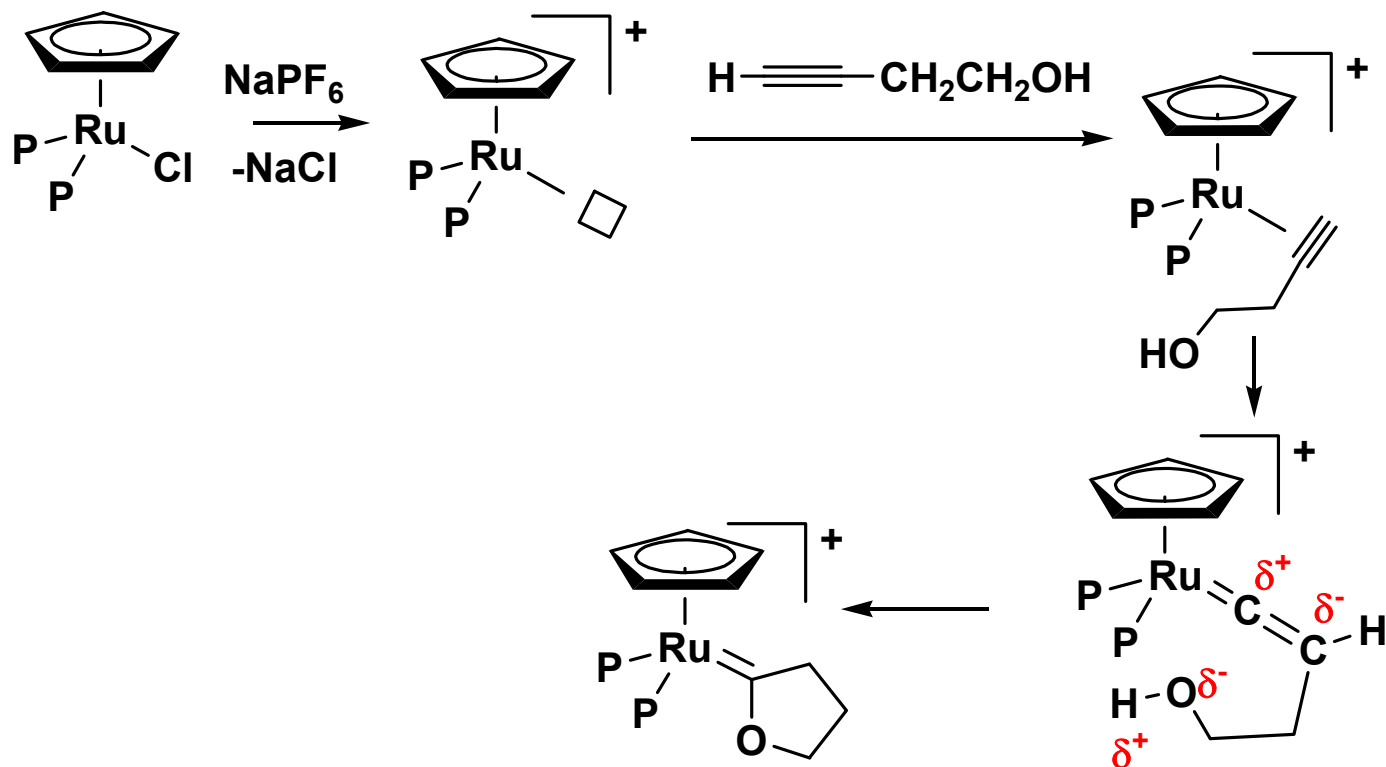


Suggest a reasonable mechanism for the following reaction



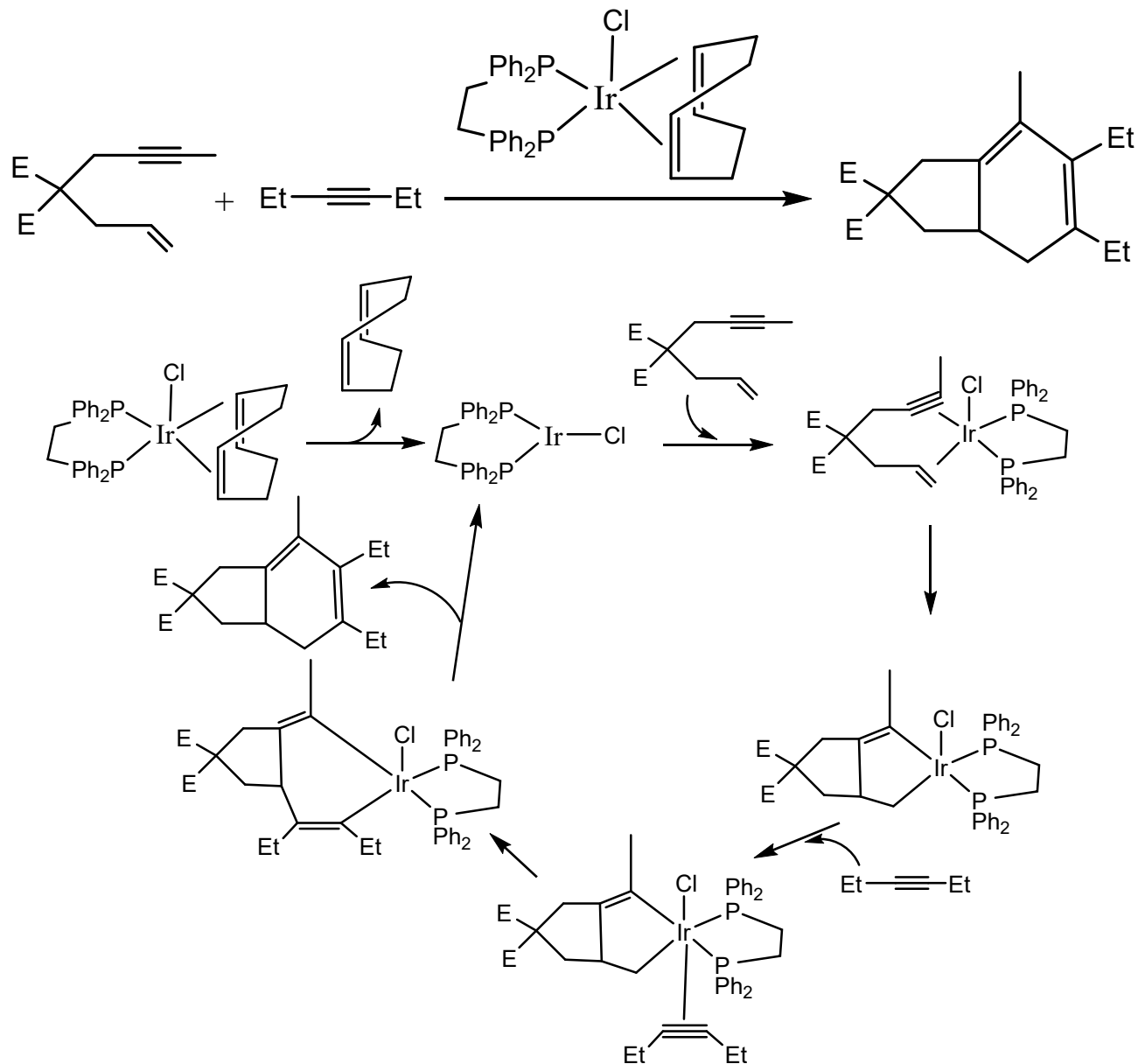


Mechanism ?

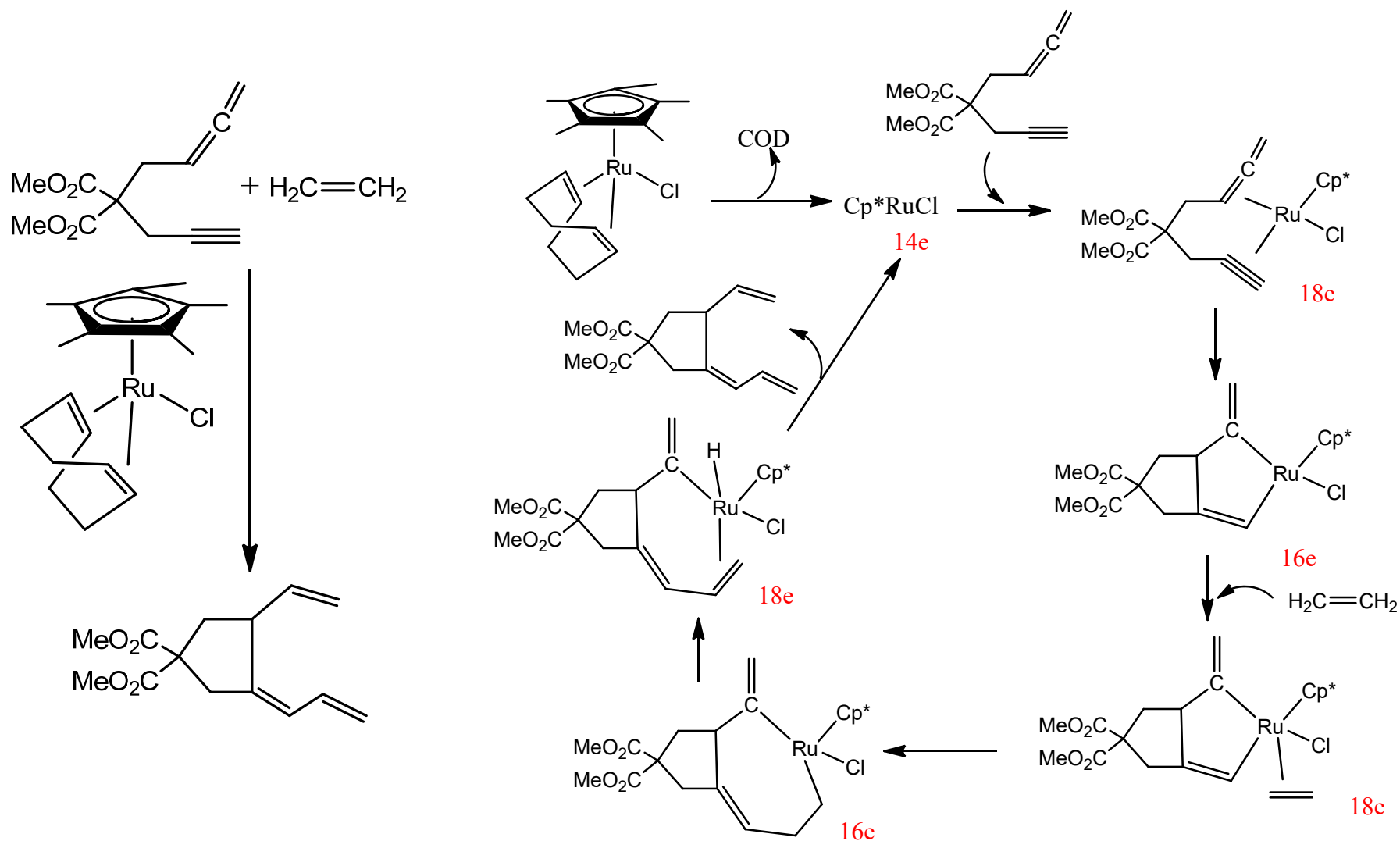


6. Mechanism for metal catalyzed reactions

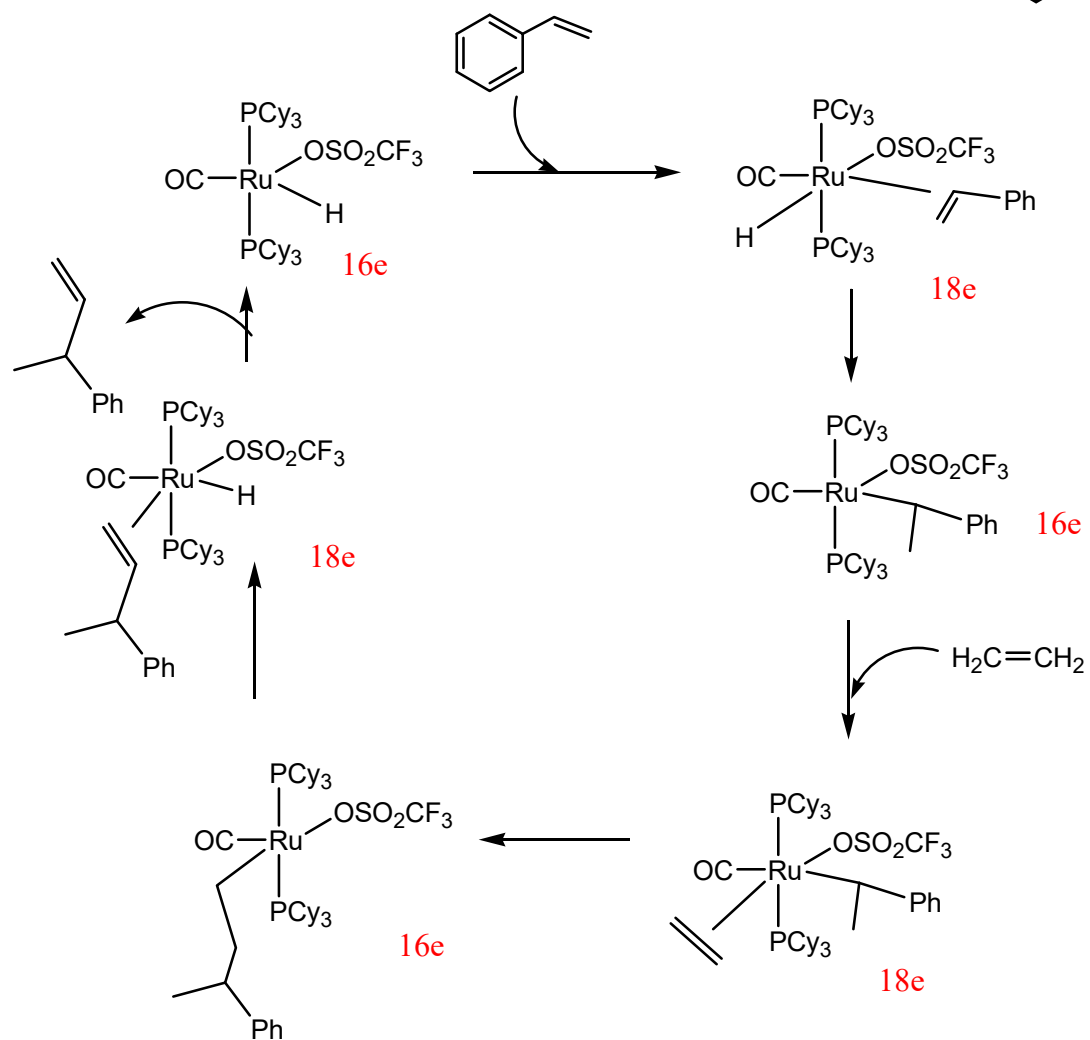
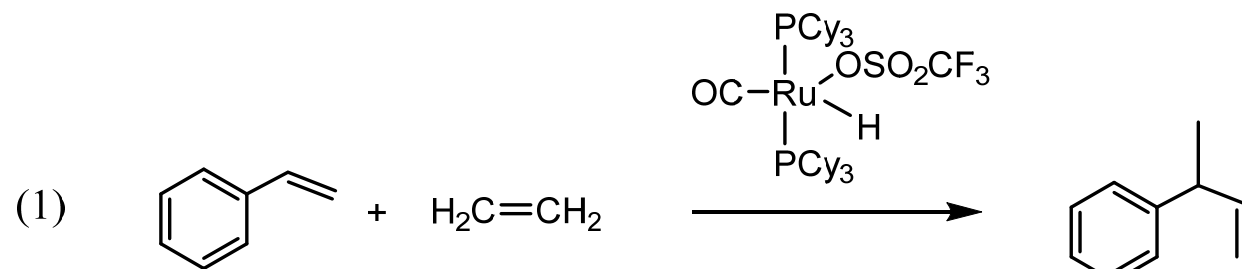
1. Propose a mechanism for the following reaction.



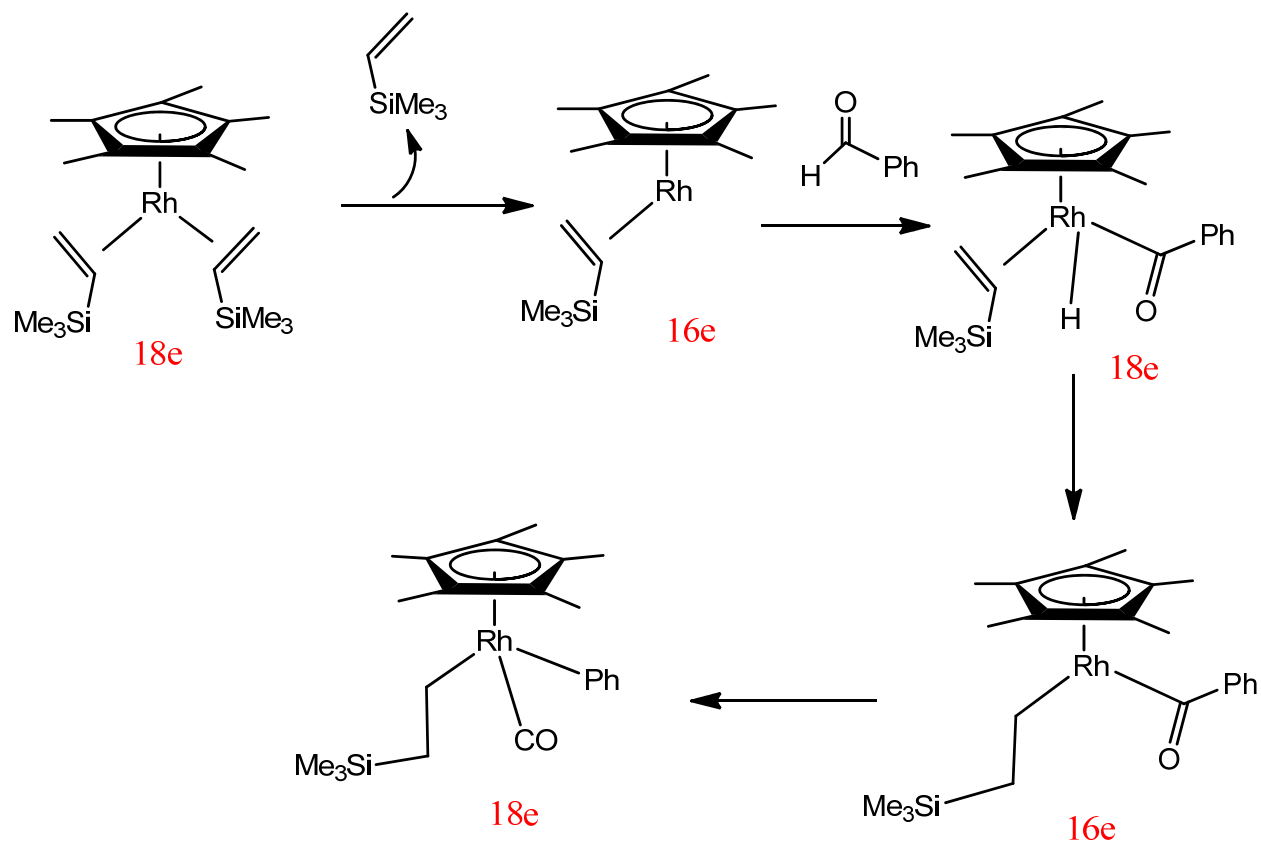
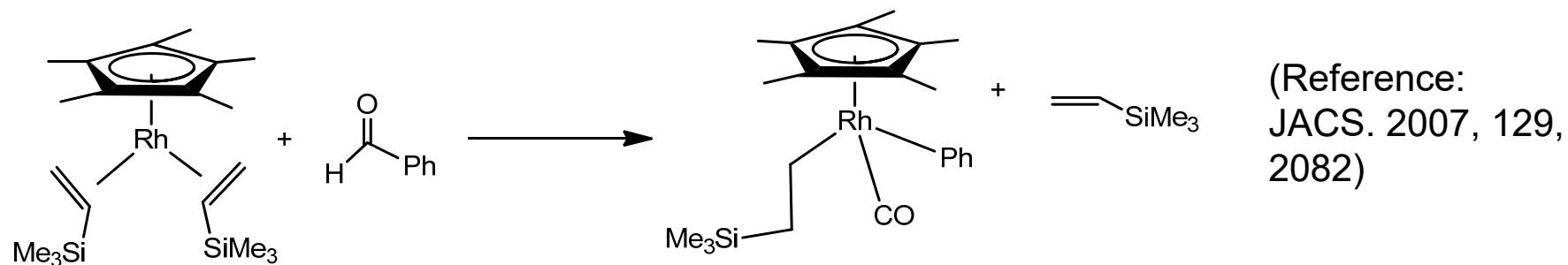
Propose a reasonable mechanism for each of the following metal catalyzed reactions.

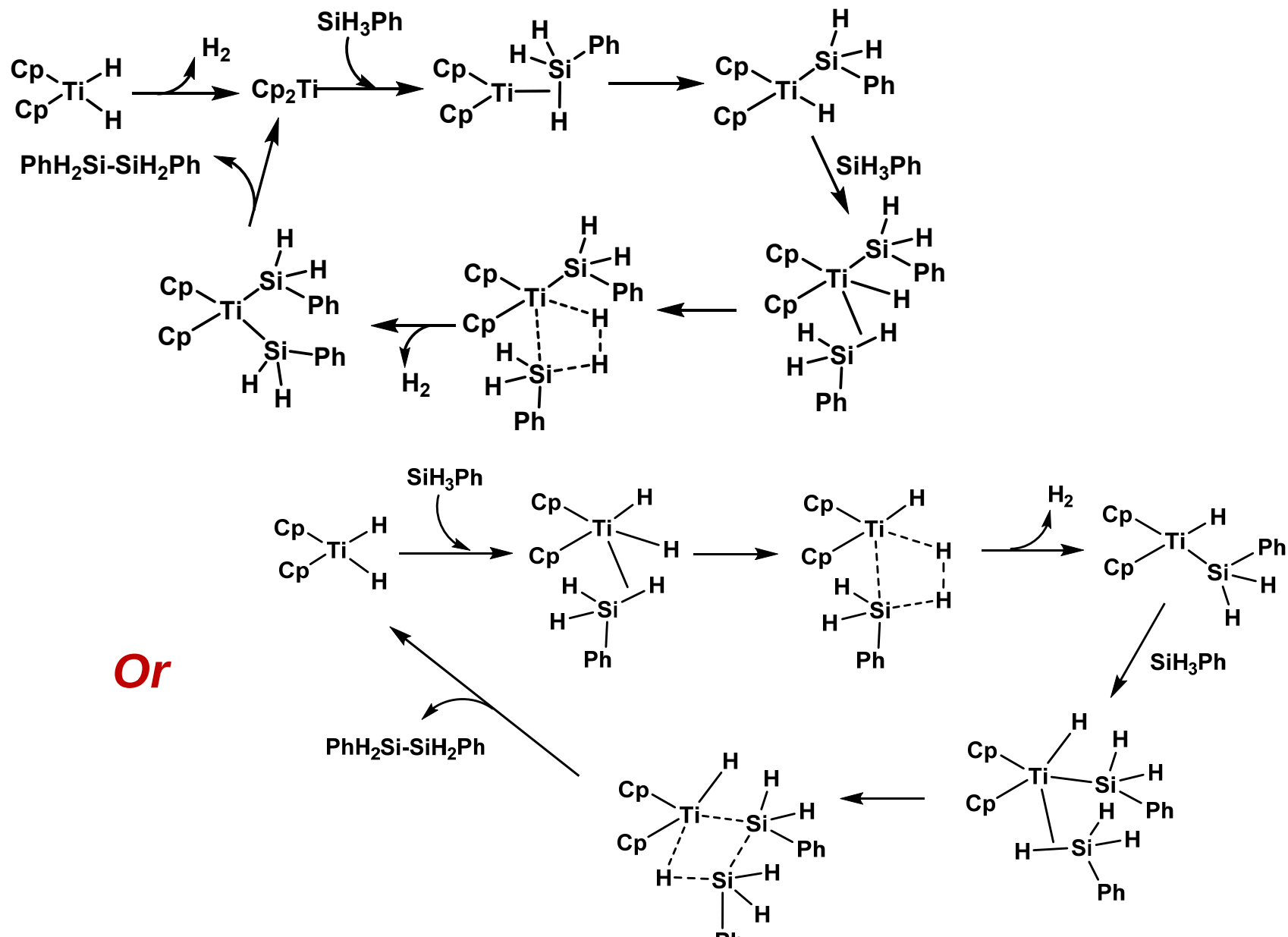
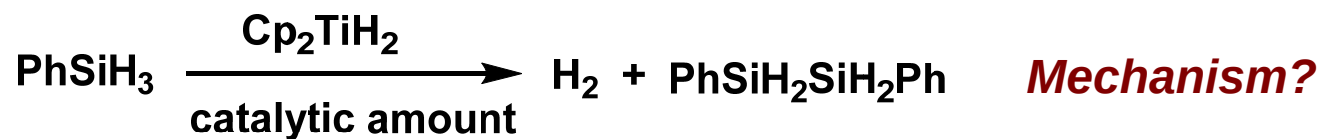


2. Suggest a reasonable mechanism for the metal catalyzed reactions.

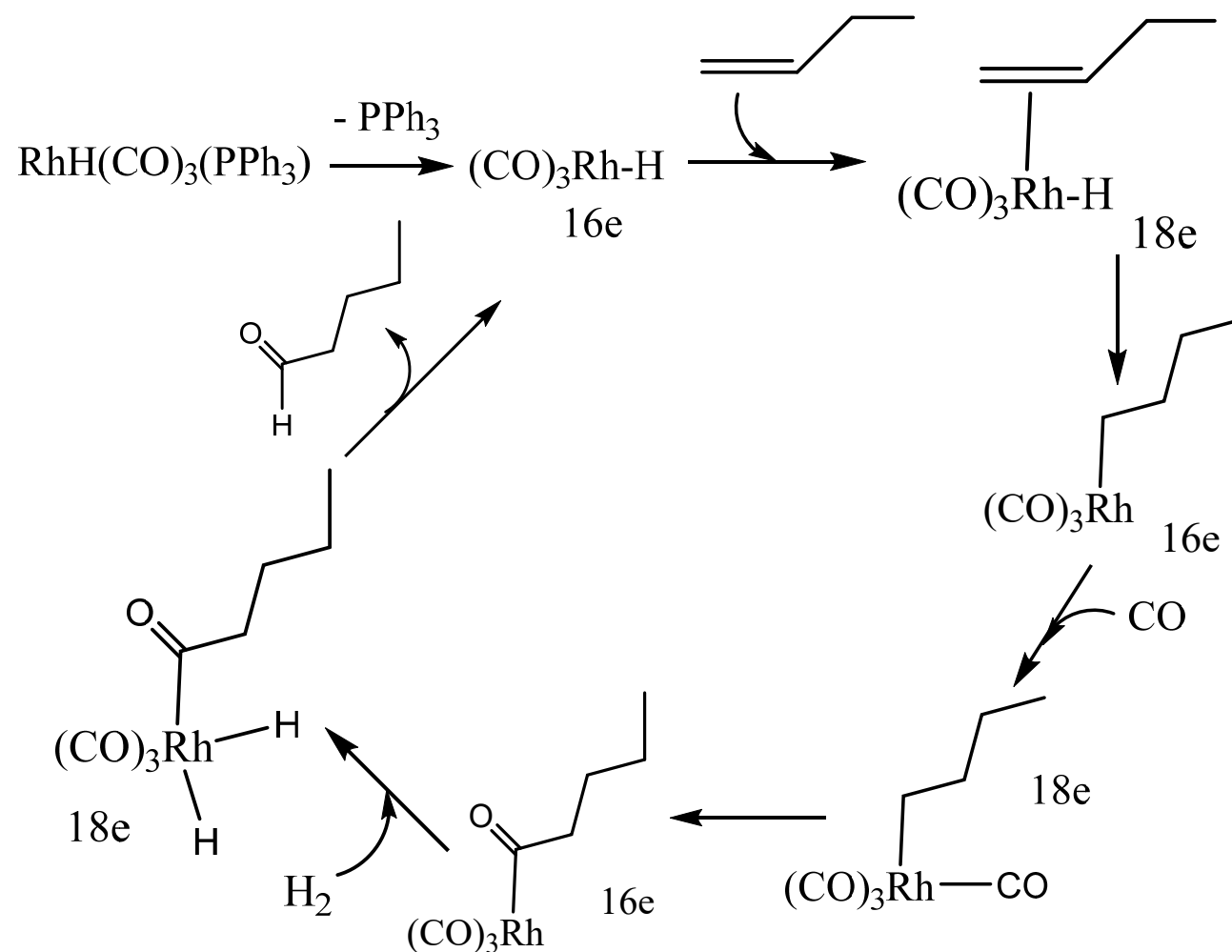
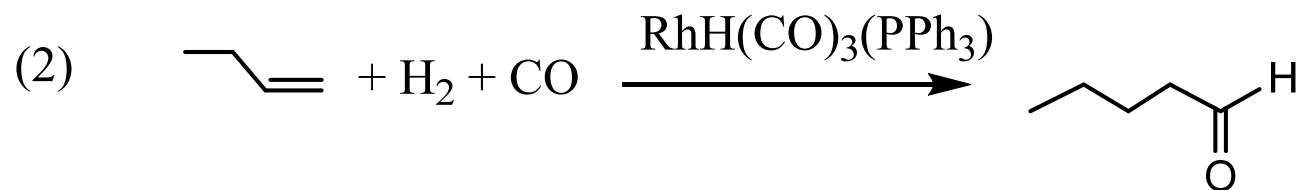


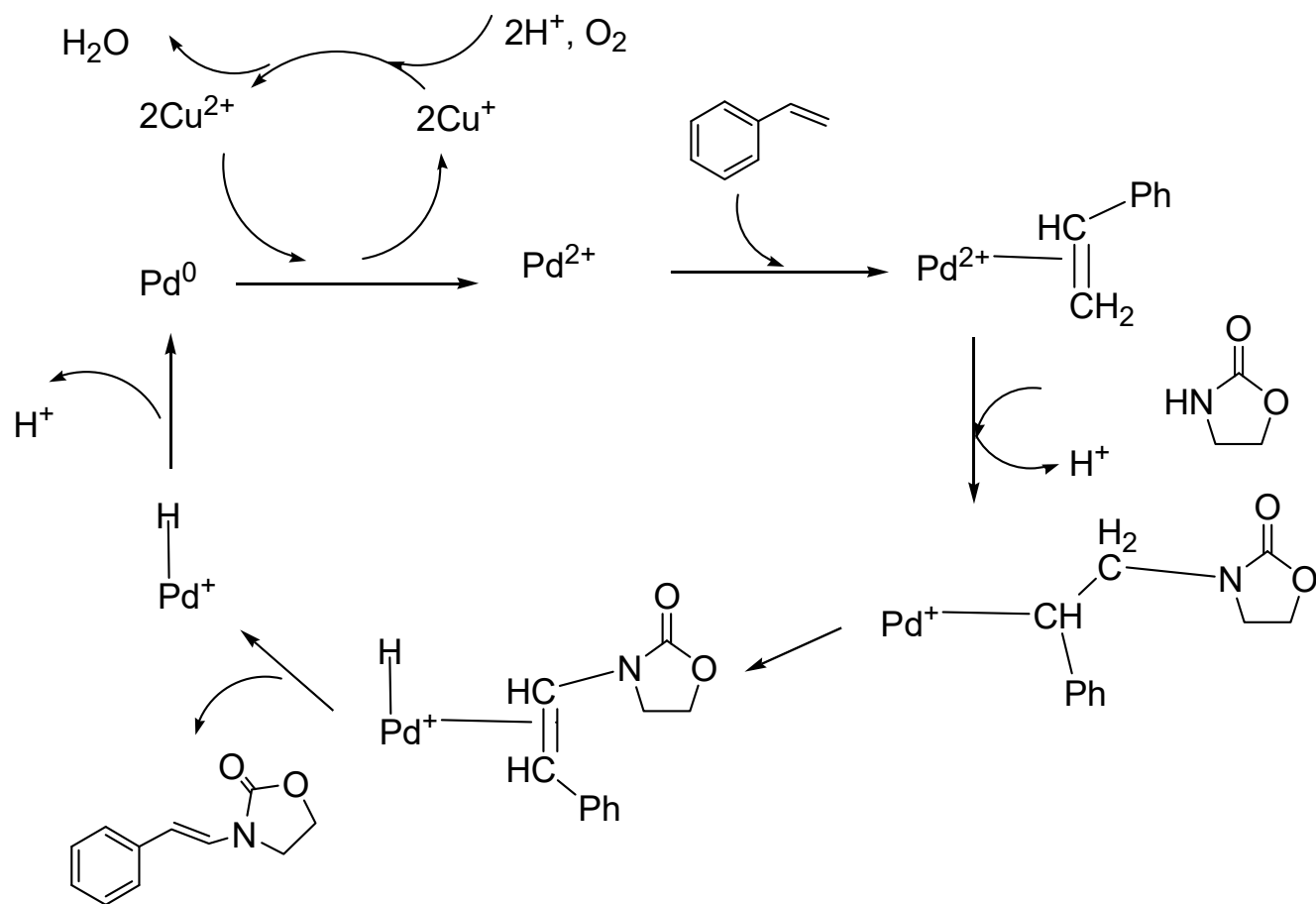
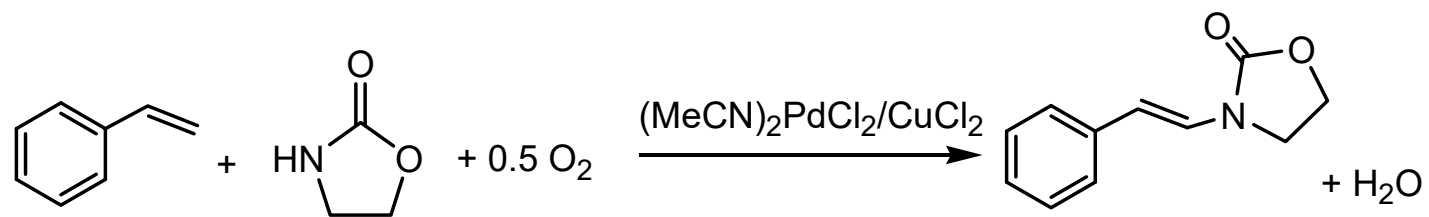
3. Suggest a reasonable mechanism for the following reaction.





7. Propose a mechanism for the following catalytic reactions.





7. Suggest a reasonable mechanism for the metal catalyzed reactions.

